

9

Liabilities



SPOTLIGHT

Jardine Matheson Holdings Ltd.

www.jardines.com



Jardine Matheson (hereafter “Jardine”) may not be a household name, but it is a large diversified conglomerate based in Hong Kong, having been founded in China in 1832. It employs over 440,000 people across varied business interests from motor vehicle and transportation, financial services and insurance, to hotels and restaurants, and even mining and agribusiness. Its subsidiaries include Hong Kong Land, Dairy Farm, Mandarin Oriental, Cycle & Carriage, and Astra International. ●

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In financial year 2015, Jardine made a net profit of \$3,982 million. To be able to do this, Jardine required a significant amount of resources, from cash, receivables, and inventory to property, plant and equipment and other assets totaling \$66,955 million. These resources are, in turn, funded by shareholders and creditors of Jardine. Shareholders' equity amounted to \$45,781 million, and creditors provided liabilities totaling \$21,174 million. Most of the liabilities are due within the next financial year, totaling \$12,602 million and the rest, \$8,572 million, are non-current. Let's see how the various types of liabilities play a role in Jardine's ability to offer goods and services to its customers around the world.

A1			
	A	B	C
1	Jardine Matheson Holdings Ltd.	As at December 31	
2	Consolidated Balance Sheet	2015	2014
3	(Adapted, in millions of US\$)		
4	Total assets	\$ 66,955	\$ 66,457
5	Current creditors	8,261	8,244
6	Current borrowings	3,991	4,068
7	Current tax liabilities	266	300
8	Current provisions	84	77
9	Total current liabilities	\$ 12,602	\$ 12,689
10	Long-term borrowings	6,995	7,416
11	Non-current creditors	430	364
12	Non-current provisions	145	138
13	Pension Liabilities	416	350
14	Other non-current liabilities	586	695
15	Total non-current liabilities	8,572	8,963
16	Total liabilities	21,174	21,652
17	Shareholders' equity	45,781	44,805
18	Total liabilities and shareholders' equity	\$ 66,955	\$ 66,457
19			

This chapter shows how to account for liabilities—both current and long-term. We begin with a refresher on liabilities and then proceed with our discussions on current liabilities.

LEARNING OBJECTIVES

1 Understand the various types of liabilities

2 Account for contingent liabilities

3 Account for bonds

4 Account for leases

5 Analyze the advantages and disadvantages of borrowing

6 Evaluate a company's debt-paying ability

UNDERSTAND THE VARIOUS TYPES OF LIABILITIES

You have seen many examples of liability accounts, but let's first recall what a liability is. The *Conceptual Framework* defines liability as a present obligation which will be settled through an outflow of resources embodying economic benefits. Liabilities are recognized on the Balance Sheet when it is probable that such an outflow will occur and the amount of which can be measured reliably. The general principle is for short-term liability to be expressed in nominal amounts and for long-term liability to be expressed in present values, taking into account the time value of money.

Liabilities are classified into current (short-term) and non-current (long-term). Current liabilities are obligations due within one year or within a company's normal operating cycle (see Chapter 4). Obligations due beyond that period of time are classified as non-current or long-term liabilities.

Lines 5 to 8 are Jardine's current liabilities (total at line 9, \$12,602 million), and lines 10 to 14 are its non-current liabilities (total at line 15, \$8,572 million). Notice that there are many similar account names in both sections, only differentiated with current versus non-current descriptions. For example, there are current and long-term borrowings, and current and long-term provisions. We will discuss liabilities by their most common headings, but remember they can be classified as either current or non-current depending on the timing of expected outflow of economic resources.

Note that Jardine has chosen to label line 5 as "current creditors." Other entities may decide to use different account names and different ways of aggregating its liabilities. For example, some may group "trade" and "other payables" together while others may list them in two separate lines. Let's continue to use Jardine's examples of current creditors in our discussion, and we will add on other types of liabilities you may see disclosed by other companies. Jardine's notes to the accounts provided the following additional information about its trade and other payable balances. We have summarized the list to include only the items we will discuss in this chapter.

1 Understand the various types of liabilities

ADAPTED EXCERPTS FROM JARDINE MATHESON'S NOTES TO THE ACCOUNTS

Creditors (in \$ millions)	2015	2014
■ Trade creditors	\$4,177	\$4,171
■ Accruals	1,586	1,626
■ Rental and other refundable deposits	392	420
■ Proceeds from properties sale received in advance	892	697
■ Rental income received in advance	20	26
■ Other income received in advance	204	214
■ All other creditors (summarized)	<u>1,420</u>	<u>1,454</u>
Total	<u>\$8,691</u>	<u>\$8,608</u>
Current	8,261	8,244
Non-current	<u>430</u>	<u>364</u>
Total	<u>\$8,691</u>	<u>\$8,608</u>

Source: Jardine Matheson's 2015 annual report, page 107

Accounts Payable. Amounts owed for products or services purchased on account are called accounts payable or trade payables or trade creditors. You have seen examples of accounts payable in preceding chapters. They are typically short-term, as credit terms are usually between 30 and 90 days. One of a merchandiser's most common transactions is the purchase of inventory, usually on credit. All businesses make the majority of their purchases on account, typically for raw materials, supplies, and inventory, as well as for various services. For example, as you can see from its Balance Sheet, Jardine's trade creditors amounted to \$4,177 million on December 31, 2015.

Accrued Liabilities (Accrued Expenses). An **accrued liability** usually results from an expense the business has incurred but not yet paid. Therefore, an **accrued expense** creates a liability, which explains why it is also called an accrued expense. Most accruals are short-term liabilities and will be shown as part of current liabilities. Jardine's accruals include wages and salary

payable, and other accruals as relevant. You have seen wages and salary payable and other accruals, such as interest payable, earlier (in Chapter 3).

Deposits. There may be occasions when a business is required to pay a deposit for securing goods, property, or serviced such as power supplies and telephony services. The deposits are refundable once the service is no longer required or when the contract lapses. Obviously, this may go both ways. Jardine may place deposits to secure services from other businesses, and in turn, Jardine’s customers may place deposits to secure Jardine’s goods or services. The former is an asset and the latter is a liability. In this case, because the deposits are shown as an item of liability, this must be the amount placed by Jardine’s customers. In fact, in its Debtors note, Jardine shows the exact same title “Rental and other refundable deposits” of \$392 million at December 31, 2015.

Unearned Revenues. Unearned revenues are also called deferred revenues or revenues collected in advance. For all unearned revenue, the business has received cash from customers before earning the revenue. The company has a liability—an obligation to provide goods or services to the customer. Jardine has a number of items related to unearned revenue: proceeds from properties for sale received in advance, rental income received in advance, and other income received in advance, totaling \$1,116 million on December 31, 2015. Unearned revenue is typically a current liability. Let’s consider another example.

Thai Airways sells tickets and collects cash in advance. Thai Airways therefore reports Unearned Ticket Revenue for airline tickets sold in advance.

Let’s consider another example that you may have experienced as an airline customer. You always pay in advance when you buy a ticket. To the airline, this is not an income (until you fly) and is recorded as unearned revenue for tickets sold in advance. Let’s see how Thai Airways would account for unearned ticket revenue. Assume that Thai Airways collects \$1,000 for a round-trip ticket from Bangkok (Thailand) to Madrid (Spain) and back. Thai Airways records the cash collection and related liability as follows:



A1				
	A	B	C	D
1	20X0			
2	Dec 15	Cash	1,000	
3		Unearned Ticket Revenue		1,000
4		Received cash in advance for ticket sales.		
5				

Unearned Ticket Revenue	
	1,000

Suppose the customer flies to Madrid late in December. Thai Airways records the revenue earned as follows:

A1				
	A	B	C	D
1				
2		Unearned Ticket Revenue	500	
3		Ticket Revenue (\$1,000 × 1/2)		500
4		Earned revenue that was collected in advance.		
5				

The liability decreases and the revenue goes up.

Unearned Ticket Revenue		Ticket Revenue	
500	1,000		500
	Bal. 500		

At the year-end, Thai Airways reports:

- \$500 of unearned ticket revenue (a liability) on the Balance Sheet; and
- \$500 of ticket revenue on the Income Statement.

The customer returns to Bangkok in January 20X1, and Thai Airways records the revenue earned with this journal entry:

A1				
	A	B	C	D
1	20X1			
2	Jan 4	Unearned Ticket Revenue	500	
3		Ticket Revenue (\$1,000 × 1/2)		500
4		<i>Earned revenue that was collected in advance.</i>		
5				

Now the liability balance is zero because Thai Airways has earned all the revenue it collected in advance.

Unearned Ticket Revenue	
500	1,000
500	
	Bal. 0

Payroll-related Liabilities. Payroll, also called employee compensation, is a substantial expense for many companies. For service organizations—such as law firms, real estate companies, and airlines—employee compensation is the major expense, just as cost of goods sold is the largest expense for a merchandising company.

Employee compensation takes many different forms. A salary is employee pay stated at a monthly or yearly rate. A wage is employee pay stated at an hourly rate. Sales employees usually earn a commission, which is a percentage of the sales the employee has made. A bonus is an amount over and above regular compensation. In some jurisdictions, employee's income tax is deducted from employee's salaries as they are paid, or the employer may be required to pay some form of payroll tax. There may also be other kinds of deductions, such as employee's contribution to a provident or superannuation fund. Salary expense thus represents gross pay (that is, employee pay before subtractions for taxes and other deductions). Your instructor may give you additional information about your local payroll regulations. Accounting for all forms of compensation follows the general pattern illustrated in Exhibit 9-1 (using assumed figures). The entries depict payment to a staff member with a monthly salary of \$5,000, employee's "pay as you earn" income tax of 20%, and an employer's contribution to superannuation of 10% on top of monthly salary.

Exhibit 9-1 Accounting for Payroll Expenses and Liabilities

A1				
	A	B	C	D
1	20X1			
2	Dec 31	Salary Expense	5,500	
3		Cash (to tax office on behalf of employee)		1,000
4		Cash (to staff)		4,000
5		Cash (to superannuation fund)		500
6		<i>Salary payment to staff, tax office and fund</i>		
7				

In countries where businesses provide pension or other post-employment benefits, they must recognize such obligations on each Balance Sheet date and record such expenses for each period. Jardine, for example, has over \$416 million of pension liabilities (line 13 on its Balance Sheet at the start of this chapter).

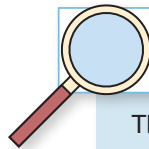
Broadly speaking, there are two basic schemes in relation to employees' post-retirement obligations: defined contributions and defined benefits. In defined contribution schemes, employers contribute a fixed amount of money to an employee's pension (also known as provident, or superannuation) funds. The employer's obligation ends once the contribution has been made. As members of the pension fund, the employees are able to use, invest, or withdraw the contribution accumulated, subject to the fund's rules and regulations.

In a defined benefit plan, the employee is promised some post-retirement benefits, usually referred to as **pensions**. Companies may also provide other post-retirement benefits, such as medical insurance for retired former employees. Because employees earn these benefits by their service, the company records pension and retirement-benefit expenses while employees work for the company.

Pensions are one of the most complex areas of accounting, and beyond the scope of this textbook. As employees earn their pensions and the company pays into the pension plan, the plan's assets grow. The obligation for future pension payments to employees also accumulates. At the end of each period, the company compares the following:

- the fair market value of the assets in the retirement plans—cash and investments
- the plans' accumulated benefit obligation, which is the present value of promised future payments to retirees

If the plan assets exceed the accumulated benefit obligation, the plan is said to be overfunded. In this case, the asset and obligation amounts are to be reported only in the notes to the financial statements. However, if the accumulated benefit obligation (the liability) exceeds plan assets, the plan is underfunded, and the company must report the excess liability amount as a long-term liability on the Balance Sheet.



A Closer Look

The accounting for pension and other post-employment benefits requires a significant amount of detailed record keeping and estimates. They are termed "defined benefits plans" because you promise your employees post-employment benefits when they retire. Imagine how you would go about recording an obligation to pay a lifelong pension to a young 25-year-old employee who just started work for you today. You promise him that when he retires at, say, 65, you will pay 25% of his annual salary for life. You don't know exactly how long he will work for you, you don't know exactly what his salary will be when he retires, and you don't know exactly how long he will live after 65! *IAS 19—Employee Benefits* provides guidance on this topic, but it is an area that is clearly beyond an introductory financial accounting textbook.

Sales Tax Payable. Most countries have some form of consumption tax. This tax is usually called Goods and Services Tax (GST), Value-Added Tax (VAT), or simply sales tax. Retailers collect the tax from customers and thus owe the tax authority for sales tax collected. Let's consider IKEA, one of Jardine's subsidiaries. Suppose one Saturday's sales at an IKEA store totaled \$200,000. IKEA collected an additional 5% (\$10,000) of sales tax. The store would record that day's sales as follows:

A1				
	A	B	C	D
1				
2		Cash ($\$200,000 \times 1.05$)	210,000	
3		Sales Revenue		200,000
4		Sales Tax Payable ($\$200,000 \times 0.05$)		10,000
5		<i>To record cash sales and the related sales tax.</i>		
6				

Assets, liabilities, and equity all increase—equity because of the sales.

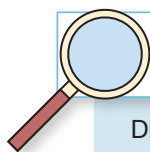
Assets	=	Liabilities	+	Shareholders' Equity	
+ 210,000	=	+ 10,000	+	+ 200,000	(sales revenue)

When the sales tax payable is remitted to the tax authority, IKEA would perform the following entry. Note that there is no impact on revenue or equity.

A1				
	A	B	C	D
1				
2		Sales tax payable	10,000	
3		Cash		10,000
4		<i>To record payment of sales tax.</i>		
5				

Assets	=	Liabilities	+	Shareholders' Equity
- 10,000	=	- 10,000	+	0

Tax Payable. Business entities are expected to pay tax on their income. Taxation rules vary from one jurisdiction to another, and in most cases, taxable income may not equal net profit. For example, in Chapter 7 we described how a company may be able to use an accelerated depreciation method for tax purposes while using straight-line depreciation for financial reporting purposes. Ignoring such differences, tax payable can be calculated as the prevailing tax rates multiplied by the profit before tax of the business. For example, Jardine's profit before taxation was \$4,591 million, and its tax expense was \$609 million, about 13.3% of pre-tax income. Depending on the tax collection schedule, corporate tax may be paid in instalments during the year (or at different times in different tax jurisdictions) and finalized at the tax submission dateline. Jardine showed a current tax payable amount of \$266 million on its Balance Sheet.



A Closer Look

Differences in taxable income and profit before tax are caused by differences between tax bases of assets and liabilities and their carrying amounts on the Balance Sheet. The tax base of an asset or liability is the amount attributed to the asset or liability for tax purposes, as opposed to its carrying amount on the Balance Sheet. These differences create deferred tax assets and deferred tax liabilities. If you are an accounting major, you are likely to be studying *IAS 12—Income Taxes* in your future courses.

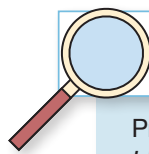
Provisions. A business may know that a liability exists but not know its exact amount. The business must still report the liability on the Balance Sheet based on the best estimates possible on the reporting date. Estimated liabilities vary among businesses.

The following adapted excerpt shows that Jardine has provisions totaling \$229 million on December 31, 2015, of which \$84 is current and \$145 is non-current. You can tally these totals to its Balance Sheet at the start of this chapter (lines 8 and 12).

ADAPTED EXCERPTS FROM JARDINE MATHESON'S NOTES TO THE ACCOUNTS

Provisions (in \$ millions)	2015 Current	2015 Non-current	2015 Total
Motor vehicle warranties	39	-	39
Closure cost provisions	7	1	8
Reinstatement and restoration costs	5	40	45
Others	<u>33</u>	<u>104</u>	<u>137</u>
Total provisions	<u>84</u>	<u>145</u>	<u>229</u>

Source: Jardine Matheson's 2015 annual report, page 108



A Closer Look

Provisions are covered under IAS 37—*Provisions, Contingent Assets and Contingent Liabilities*. They are defined as liabilities of uncertain timing and amount. In other words, they meet the definition of a liability. Sometimes you see businesses use the word “provision” to describe an allowance, for example “provision for bad debts” or “provision for inventory obsolescence.” This is technically incorrect.

Many businesses, especially manufacturers, guarantee their products under some form of warranty agreement. The warranty period may extend from a short period (e.g., 30 days to 12 months warranty for consumer products) to multiple years for major assets (such as bridges, airplanes, and other infrastructure assets). For example, automobile companies—Peugeot Citroën, BMW, and Toyota—all make provisions for vehicle warranties.

Whatever the warranty's life, businesses need to recognize that they have an obligation to perform services or repairs should the need arise. This increase in liability results in a decrease in equity, and thus an expense (usually called warranty expense) is created and recognized in each reporting period. At the time of the sale, however, the company doesn't know which products are (or will be) defective. The exact amount of warranty expense cannot be known with certainty, so the business must estimate warranty expense and the related liability.

Assume that Black & Decker Corporation, which manufactures power tools, made sales of 100,000 power tools subject to product warranties in a financial year. Black and Decker has information based on product testing and historical experience that 1% of the power tools may require warranty service costing an average of \$50 within the warranty period, and a further 0.25% may require warranty service costing an average of \$100. Black and Decker could use the “probability-weighted expected value” in IAS 37 to work out that it has a warranty obligation amounting to \$75,000 ($1\% \times 100,000 \text{ units} \times \$50 + 0.25\% \times 100,000 \text{ units} \times \100). In this case, Black and Decker would recognize a warranty expense of \$75,000 and make the following entry:



A1				
	A	B	C	D
1		Warranty Expense	75,000	
2		Provision for Warranty Repairs		75,000
3		To accrue warranty expense.		
4				

Provision for Warranty Repairs
75,000

Warranty Expense
75,000

If in the following year, Black and Decker actually spent \$72,000 repairing these faulty products (assuming it took spare parts or replacement units from its inventory), it would then record the following:

A1				
	A	B	C	D
1		Provision for Warranty Repairs	72,000	
2		Inventory		72,000
3		<i>To replace defective products sold under warranty.</i>		
4				

Provision for Warranty Repairs

72,000	75,000
	Bal. 3,000

At the end of the year, Black and Decker will repeat the process. It starts by re-estimating the required amount of provision, compares it to the current level of provision, and the difference would be the warranty expense for the year. The Estimated Warranty Payable account probably won't ever zero out. And obviously in real life, product sales (and warranty repairs) often happen within the same period.

Notes Payable. Notes are a common form of borrowing in some parts of the world. Depending on the maturity or term of the note, you may have short-term notes (due within one year) or occasionally medium-term notes, usually up to two or three years but they can be longer as well. We will discuss long-term notes payable together with bonds in the next section.

For example, a business may issue **short-term notes payable** to borrow cash or to purchase assets. On its notes payable, the business must accrue interest expense and interest payable at the end of the period. The following sequence of entries covers the purchase of inventory, accrual of interest expense, and payment of a 10% short-term note payable that's due in one year.

A1				
	A	B	C	D
1	20X0			
2	Mar 1	Inventory	8,000	
3		Note Payable, Short-Term		8,000
4		<i>Purchase of inventory by issuing a note payable.</i>		
5				

This transaction increases both an asset and a liability.

Assets	=	Liabilities	+	Shareholders' Equity
+ 8,000	=	+ 8,000	+	0

At year-end, the business must accrue interest expense at 10% for March through December:

A1				
	A	B	C	D
1	Dec 31	Interest Expense ($\$8,000 \times 0.10 \times 9/12$)	600	
2		Interest Payable		600
3		<i>Accrual of interest expense at year-end.</i>		
4				

Liabilities increase and equity decreases because of the expense.

Assets	=	Liabilities	+	Shareholders' Equity	
0	=	+ 600	+	– 600	(interest expense)

The Balance Sheet at year-end will report the Note Payable of \$8,000 and the related Interest Payable of \$600 as current liabilities. The Income Statement will report Interest Expense of \$600.

The following entry records the note's payment at maturity on March 1, 20X1:

A1				
	A	B	C	D
1	20X1			
2	Mar 1	Note Payable, Short-Term	8,000	
3		Interest Payable	600	
4		Interest Expense ($\$8,000 \times 0.10 \times 3/12$)	200	
5		Cash [$\$8,000 + (\$8,000 \times 0.10)$]		8,800
6		Payment of a note payable and interest at maturity.		
7				

The debits zero out the payables and also record the business's interest expense for January, February, and March 20X1.

Debt. There are many forms of borrowing, from a simple term loan to a complex financing structure. Some long-term debt must be paid in installments. Some only require interest payments during the term of the loan, with payment of the loan amount at a specific time in the future. Whatever the case may be, the part of the loan that is due in the coming financial year is called **current portion of long-term debt** and is shown as a current liability. At the end of each year, an entity reclassifies (from long-term debt to a current liability) the amount of its long-term debt that must be paid the next year.

Jardine's Balance Sheet at the start of this chapter showed that its current or short-term borrowings is at \$3,991 million (line 6) and long-term borrowings is at \$6,995 million (line 10). Long-term debt refers to long-term notes payable and bonds payable, which we will cover in the later part of this chapter.

You have seen the broad categories of liabilities, both current and non-current. Now, let's look at something that sounds like a liability but is actually not a liability.

ACCOUNT FOR CONTINGENT LIABILITIES

2 Account for contingent liabilities

A contingent liability is *not* an actual liability. A contingent liability is a disclosure item in the notes to the financial statement. IAS 37 states that a contingent liability arises when:

- there is a possible obligation to be confirmed by a future event that is outside the control of the entity; or
- a present obligation may, but probably will not, require an outflow of resources; or
- a sufficiently reliable estimate of the amount of a present obligation cannot be made.

Note that in each instance, contingent liabilities fail to meet one aspect of the definition of liability. Examples of contingent liabilities include corporate guarantees, lawsuits, tax disputes, or alleged violations of environmental protection laws. If, for example, the probability of a particular contingent liability has risen from possible to probable, the company would have to make a provision (i.e., recognize an expense and a related liability) in anticipation of the settlement of such liability. In some industries, contingent liabilities may be more prevalent than others. Let's look at the contingent liability disclosure of BP, especially in relation to the *Deep Horizon* (Gulf of Mexico) oil spill in 2010.

ADAPTED EXCERPTS FROM BP's NOTES TO THE ACCOUNTS

Contingent liabilities are possible obligations whose existence will only be confirmed by future events not wholly within the control of the group, or present obligations where it is not probable that an outflow of resources will be required or the amount of the obligation cannot be measured with sufficient reliability. Contingent liabilities are not recognized in the financial statements but are disclosed unless the possibility of an outflow of economic resources is considered remote.

Gulf of Mexico Oil Spill

As a consequence of the Gulf of Mexico oil spill in April 2010, BP continues to incur costs and has also recognized liabilities for certain future costs. The cumulative pre-tax Income Statement charge since the incident amounts to \$55.5 billion. The cumulative Income Statement charge does not include amounts for obligations that BP considers are not possible, at this time, to measure reliably. BP has recorded provisions [\$16.5 billion] relating to the Gulf of Mexico oil spill in relation to environmental expenditure (including spill response costs), litigation and claims, and Clean Water Act penalties that can be measured reliably at this time.

BP has provided for its best estimate of amounts expected to be paid that can be measured reliably. It is not possible, at this time, to measure reliably other obligations arising from the incident, nor is it practicable to estimate their magnitude or possible timing of payment. Therefore, no amounts have been provided for these obligations at 31 December 2015.

Other Contingent Liabilities

In the normal course of the group's business, legal proceedings are pending or may be brought against BP group entities arising out of current and past operations, including matters related to commercial disputes, product liability, antitrust, commodities trading, premises-liability claims, consumer protection, general environmental claims and allegations of exposures of third parties to toxic substances, such as lead pigment in paint, asbestos and other chemicals. BP believes that the impact of these legal proceedings on the group's results of operations, liquidity or financial position will not be material.

Source: BP's 2015 annual report, pages 114, 117, 119, 120, 159–160

It is important to remember that there is no need to report a contingent loss that is unlikely to occur. Instead, an entity should wait until new information is available to provide further clarity on the situation. For example, suppose Del Monte Foods grows vegetables in a foreign country, where the new president has made threats that it will nationalize (i.e., will confiscate the assets of all foreign companies) Del Monte's plantation. Del Monte will report nothing about the contingency if the probability of such an eventuality is considered remote.



Are All Liabilities Reported on the Balance Sheet?

The big danger with liabilities is that you may fail to report a large debt on your Balance Sheet. What is the consequence of missing a large liability? You will definitely understate your liabilities and your debt ratio (the percentage of your assets financed by liabilities—more on this later in the chapter). By failing to accrue interest on the liability, you'll probably overstate your net income as well. In short, your financial statements will make you look better than you really are. Any such error, if significant, hurts a company's credibility.

Contingent liabilities are very easy to overlook because they aren't actual debts. How would you feel if you owned shares in a company that failed to report a contingency that eventually caused the company to go out of business? If you had known of the contingency, you could have sold the shares and avoided the loss.



COOKING THE BOOKS

with Liabilities

Crazy Eddie, Inc.

Accidentally understating liabilities is one thing, but doing it intentionally is quite another. When unethical management decides to cook the books in the area of liabilities, its strategy is to deliberately understate recorded liabilities. This can be done by intentionally under-recording the amount of existing liabilities, or by omitting certain liabilities altogether.

Crazy Eddie, Inc. (first discussed in Chapter 6) used multiple tactics to overstate its financial position from 1984 through 1987. In addition to overstating inventory (thus understating cost of goods sold and overstating income), the management of the company deliberately understated accounts payable by issuing fictitious (false) debit memoranda from suppliers (vendors). A debit memo is issued for goods returned to a vendor, such as Sony. When a debit memorandum is issued, accounts payable are debited (reduced), thus reducing current liabilities and increasing the current ratio. Eventually, expenses are also decreased, and profits are correspondingly increased through reduction of expenses. Crazy Eddie, Inc., issued \$3 million of fictitious debit memoranda in 1985, making the company's current ratio and debt ratio look better than they actually were, and eventually overstating profits.

SUMMARY OF CURRENT LIABILITIES

Let's summarize what we've covered thus far. A company can report its current liabilities on the Balance Sheet as follows:

A1			
	A	B	C
	Accounting, Inc. Balance Sheet December 31, 20X0		
1			
2	Assets		Liabilities
3	Current Assets:		Current liabilities:
4	Cash		Accounts payable
5	Short-term investments		Salary payable*
6	Etc.		Interest payable*
7			Income tax payable*
8	Non-current Assets:		Unearned revenue
9	PPE		Provision for warranty repairs*
10	Intangibles		Notes payable, short-term
11	Etc.		Current portion of long-term debt
12			Total current liabilities
13			Long-term liabilities
14			
15			Shareholders' Equity
16			Share capital
17			Retained earnings
18			
19			Total liabilities and
20	Total assets	\$ XXX	shareholders' equity
21			\$ XXX

*These items are often combined and reported in a single total as "Accrued liabilities" or "Accrued expenses payable."

Mid-Chapter

Summary Problem

Assume that Estée Lauder faced the following liability situations at June 30, 20X1, the end of the company's fiscal year. Show how Estée Lauder would report these liabilities on its Balance Sheet at June 30, 20X1.

- Salary expense for the last payroll period of the year was \$900,000. Of this amount, employees' withheld income tax totaled \$88,000 and employer's payroll taxes were \$61,000. These payroll amounts will be paid in early July.
- In fiscal year 20X1, management estimates a new warranty obligation of \$8 million arising from sales in 20X1. One year ago, at June 30, 20X0, provision for warranty stood at \$3 million. Warranty payments were \$9 million during the year ended June 30, 20X1.
- The company pays royalties on its purchased trademarks. Royalties for the trademarks are equal to a percentage of Estée Lauder's sales. Assume that sales in 20X1 were \$400 million and were subject to a royalty rate of 3%. At June 30, 20X1, Estée Lauder owes two-thirds of the year's royalty, to be paid in July.
- Long-term debt totals \$100 million and is payable in annual installments of \$10 million each. The interest rate on the debt is 7%, and the interest is paid each December 31.

Answer

Liabilities at June 30, 20X1:	
a. Current liabilities:	
Salary payable (\$900,000 – \$88,000 – \$61,000)	\$ 751,000
Employee income tax payable	88,000
Employer payroll tax payable.....	61,000
b. Current liabilities:	
Estimated warranty payable.....	2,000,000
[\$3,000,000 – \$9,000,000 + \$8,000,000]	
c. Current liabilities:	
Royalties payable ($\$400,000,000 \times 0.03 \times 2/3$)	8,000,000
d. Current liabilities:	
Current installment of long-term debt	10,000,000
Interest payable ($\$100,000,000 \times 0.07 \times 6/12$)	3,500,000
Long-term debt ($\$100,000,000 - \$10,000,000$).....	90,000,000

ACCOUNT FOR BONDS

Let's start by having a look at Jardine's non-current borrowings (line 10) on its Balance Sheet at the start of this chapter. The \$6,995 non-current borrowings are comprised of the following.

3 Account for bonds

ADAPTED EXCERPTS FROM JARDINE MATHESON'S NOTES TO THE ACCOUNTS

Non-current Borrowings (USD millions)	2015	2014
■ Bank loans	\$2,916	\$3,448
■ Bonds and notes	4,009	3,914
■ Finance lease liabilities	65	48
■ Other loans	5	6
	<u>6,995</u>	<u>7,416</u>

Source: Jardine Matheson's 2015 annual report, page 102

The biggest source of Jardine's non-current borrowing is bonds and notes. In fact, Jardine has over 50 bonds and notes outstanding. In this section, we will focus on bonds and notes, and we will look at finance lease later. Here we treat bonds payable and notes payable together because their underlying accounting treatment is similar. The main difference is that bonds tend to be of longer duration than notes, and even this varies from issuer to issuer.

Large companies such as Unilever, Procter & Gamble, and Jardine cannot borrow billions of dollars from a single lender. So how do corporations borrow huge amounts? They issue (sell) bonds to the public. **Bonds payable** are groups of notes payable issued to multiple lenders, called bondholders. Jardine needs financing for its operations and can borrow large amounts by issuing bonds or notes payable to thousands of individual investors, who each lend Jardine a modest amount. Jardine receives the cash it needs, and individual investors limit risk by diversifying their funds.

Don't assume that bonds and notes are limited to companies. In fact, bonds were originally used by governments or public authorities to raise long-term funds well before they were used by companies as a source of funding. The first official government bond was issued by the Bank of England in 1693 to fund a war!

Bonds: An Introduction

Each bond payable is, in effect, a note payable. Bonds payable are debts of the issuing company.

Purchasers of bonds receive a bond's certificate, which carries the issuing company's name. The certificate also states the principal, which is typically stated in units of \$1,000; principal is also called the bond's face value, maturity value, or par value. The bond obligates the issuing company to pay the debt at a specific future time called the maturity date.

Interest expense is the rental fee on borrowed money. The bond certificate states the interest rate that the issuer will pay the holder and the dates that the interest payments are due (generally twice a year). Exhibit 9-2 shows an actual bond certificate.

Issuing bonds usually requires the services of a securities firm, such as the Bank of America, to act as the underwriter of the bond issue. The **underwriter** purchases the bonds from the issuing company and resells them to its clients, or it may sell the bonds to its clients and earn a commission on the sale.

Types of Bonds. All the bonds in a particular issue may mature at the same time (**term bonds**) or in installments over a period of time (**serial bonds**). Serial bonds are like installment notes payable. Some long-term debts are serial in nature because they are payable in installments.

Secured, or mortgage, bonds give the bondholder the right to take specified assets of the issuer if the company defaults—that is, fails to pay interest or principal. Unsecured bonds, called **debentures**, are backed only by the good faith of the borrower. Debentures carry a higher rate of interest than secured bonds because debentures are riskier investments.

Bond Prices. Investors may buy and sell bonds through bond markets. Bond prices are quoted at a percentage of their maturity value. For example:

- A \$1,000 bond quoted at 100 is bought or sold for \$1,000, which is 100% of its face value.
- The same bond quoted at 101.5 has a market price of \$1,015 (101.5% of face value = $\$1,000 \times 1.015$).
- A \$1,000 bond quoted at 88.375 is priced at \$883.75 ($\$1,000 \times 0.88375$).

While the price of a bond on the market may change, it is important to remember that the face value and interest payments do not change. The accounting for bonds from the issuer perspective ignores fluctuations of market price, but it is disclosed in the notes.

Bond Premium and Bond Discount. A bond issued at a price above its face (par) value is said to be issued at a **premium**, and a bond issued at a price below face (par) value has a **discount**.

Premium on Bonds Payable has a credit balance and Discount on Bonds Payable carries a debit balance. Bond Discount is therefore a contra liability account.

As a bond nears maturity, its market price moves toward par value. Therefore, the price of a bond issued at a:

- premium decreases towards face value;
- discount increases towards face value.

Exhibit 9-2 Bond Certificate (Adapted)

Issuing corporation → **WASHINGTON PUBLIC POWER SUPPLY SYSTEM**

Maturity date → **ON THE FIRST DAY OF JULY 2015**

Annual interest rate → **SIX AND ONE-HALF PER CENTUM (6 1/2%)**

Semi-annual interest payments on January 1 and July 1 of each year → **ON THE FIRST DAY OF JULY 2015**

Principal amount → **FIVE THOUSAND DOLLARS (\$5,000)**

On the maturity date, a bond's market value exactly equals its face value because the company that issued the bond pays that amount to retire the bond.

The Time Value of Money. A dollar received today is worth more than a dollar to be received in the future. You can invest today's dollar immediately and earn income from it. But if you must wait to receive the dollar, you forgo the interest revenue. Money earns income over time; we call this "time value of money" (see Appendix B, available on www.pearsonglobaleditions.com/Harrison). Let's examine how the time value of money affects the pricing of bonds.

Assume that you are looking at a Jardine bond with a face value of \$1,000 that reaches maturity three years from today and carries no interest. Would you pay \$1,000 today to purchase this bond? No, because the payment of \$1,000 today to receive the same amount in the future provides you with no income on the investment. Just how much would you pay today to receive \$1,000 at the end of three years? The answer is an amount less than \$1,000. Let's suppose that you feel \$750 is a good price. By investing \$750 now to receive \$1,000 later, you earn \$250 interest revenue over the three years. The issuing company, such as Jardine, sees the transaction this way: Jardine will pay you \$250 interest to use your \$750 for three years.

The amount to invest now to receive more later is called the **present value** of a future amount. In our example, \$750 is the present value, and \$1,000 is the future amount.

Our \$750 bond price is a reasonable estimate. The exact present value of any future amount depends on:

- the amount of the future payment (\$1,000 in our example);
- the length of time from the investment date to the date when the future amount is to be collected (three years);
- the interest rate during the period (say 10%).

In this case the present value is very close to \$750. Present value is always less than the future amount. We discuss how present value is computed in Appendix B (available on www.pearson-globaleditions.com/Harrison).

Bond Interest Rates Determine Bond Prices. Bonds are always sold at their market price, which is the amount investors will pay for the bond. Market price is the bond’s present value, which equals the present value of the principal payment plus the present value of the cash interest payments. Interest is usually paid semi-annually (twice a year). Some companies pay interest annually or quarterly.

Two interest rates work to set the price of a bond:

1. The **stated interest rate**, also called the coupon rate, is the interest rate printed on the bond certificate. The stated interest rate determines the amount of cash interest the borrower pays—and the investor receives—each year. Suppose GSK bonds have a stated interest rate of 9%. GSK would pay \$9,000 of interest annually on each \$100,000 bond. Each semi-annual payment would be \$4,500 ($\$100,000 \times 0.09 \times 6/12$).
2. The **market interest rate**, or effective interest rate, is the rate that investors demand for loaning their money. The market interest rate can fluctuate after issuance of a bond.

A company may issue bonds with a stated interest rate that differs from the prevailing market interest rate. In fact, the two interest rates often differ.

Exhibit 9-3 shows how the stated interest rate and the market interest rate interact to determine the issue price of a bond payable for three separate cases.

Jardine may issue 9% bonds when the market rate is at 10%. Will Jardine’s 9% bonds attract investors in this market? No, because investors can earn 10% on other bonds of similar risk. Therefore, investors will purchase Jardine bonds only at a price less than their face value. The difference between the lower price and face value is a discount (Exhibit 9-3). Conversely, if the

Exhibit 9-3 | How the Stated Interest Rate and the Market Interest Rate Interact to Determine the Price of a Bond

Issue Price of Bonds Payable				
Case A:				
Stated interest rate on a bond payable	equals	Market interest rate	Therefore,	Price of face (par, or maturity) value
Example: 9%	=	9%	→	Par: \$1,000 bond issued for \$1,000
Case B:				
Stated interest rate on a bond payable	less than	Market interest rate	Therefore,	Discount price (price below face value)
Example: 9%	<	10%	→	Discount: \$1,000 bond issued for a price below \$1,000
Case C:				
Stated interest rate on a bond payable	greater than	Market interest rate	Therefore,	Premium price (price above face value)
Example: 9%	>	8%	→	Premium: \$1,000 bond issued for a price above \$1,000

market interest rate is 8%, Jardine's 9% bonds will be so attractive that investors will pay more than face value to purchase them. The difference between the higher price and the face value is a premium.

Issuing Bonds Payable at Par (Face Value)

We start with the most straightforward situation—issuing bonds at their par value. There is no premium or discount on these bonds payable.

Suppose Jardine issues \$50,000, 9% bonds at par on January 1, 20X1. The bonds pay interests on a semi-annual basis (July 1 and January 1 each year following issuance) and are of five years' duration, i.e., they will mature on January 1, 20X6. The issuance entry is:

A1				
	A	B	C	D
1	20X1			
2	Jan 1	Cash	50,000	
3		Bonds Payable		50,000
4		<i>To issue bonds at par.</i>		
5				

Bonds Payable	
	50,000

Assets and liabilities increase when a company issues bonds payable.

Assets	=	Liabilities	+	Shareholders' Equity
+ 50,000	=	+ 50,000	+	0

Jardine, the borrower, makes a one-time entry to record the receipt of cash and the issuance of bonds. Afterward, investors buy and sell the bonds through the bond markets. These later buy-and-sell transactions between outside investors do not involve Jardine at all.

Interest payments occur each January 1 and July 1. Jardine's entry to record the first semi-annual interest payment is:

A1				
	A	B	C	D
1	20X1			
2	Jul 1	Interest Expense ($\$50,000 \times 0.09 \times 6/12$)	2,250	
3		Cash		2,250
4		<i>To pay semi-annual interest.</i>		
5				

The payment of interest expense decreases assets and equity. Bonds payable are not affected.

Assets	=	Liabilities	+	Shareholders' Equity
- 2,250	=	0	+	- 2,250 (interest expense)

At its financial year-end, Jardine accrues interest expense and interest payable for six months (July through December), as follows:

A1				
	A	B	C	D
1	20X1			
2	Dec 31	Interest Expense ($\$50,000 \times 0.09 \times 6/12$)	2,250	
3		Interest Payable		2,250
4		<i>To accrue interest.</i>		
5				

Liabilities increase, and equity decreases.

Assets	=	Liabilities	+	Shareholders' Equity	
0	=	+2,250	+	− 2,250	(interest expense)

On January 1, Jardine will pay the interest, debiting Interest Payable and crediting Cash. Then, at maturity, Jardine pays off the bonds as follows:

A1				
	A	B	C	D
1	20X6			
2	Jan 1	Bonds Payable	50,000	
3		Cash		50,000
4		<i>To pay bonds payable at maturity.</i>		
5				

Bonds Payable	
50,000	50,000
Bal.	0

Assets	=	Liabilities	+	Shareholders' Equity
− 50,000	=	− 50,000	+	0

Issuing Bonds Payable at a Discount

Market conditions may force a company to issue bonds at a discount. Suppose Jardine issues \$100,000 of 9%, five-year bonds when the market interest rate is 10%. The issuance price of the bonds drops, and Jardine receives \$96,139¹ at issuance. The transaction is recorded as follows:

¹Appendix B (available on www.pearsonglobaleditions.com/Harrison) shows how to determine the price of this bond. If you are using Excel, you can use the PV function, i.e., PV (market rate per period, number of periods, interest payment per period and face value) or “= PV (5%, 10, −4,500, −100,000)” to get \$96,139. Any difference from workings using Appendix B is due to rounding of time value factors.

A1				
	A	B	C	D
1	20X1			
2	Jan 1	Cash	96,139	
3		Discount on Bonds Payable	3,861	
4		Bonds Payable		100,000
5		<i>To issue bonds at a discount.</i>		
6				

The accounting equation shows that Jardine has a net liability of \$96,139—not \$100,000.

Assets	=	Liabilities	+	Shareholders' Equity
+ 96,139	=	- 3,861	+	0
		+ 100,000		

The bonds payable accounts have a net balance of \$96,149 as follows:

Bonds Payable		Discount on Bonds Payable		Net carrying amount of bonds payable
100,000	-	3,861	=	\$96,139

Jardine's Balance Sheet immediately after issuance of the bonds would report the following:

Non-current liabilities:			
Bonds payable, 9%, due 2015.....	\$100,000		
Less: Discount on bonds payable.....	(3,861)	96,139	

Discount on Bonds Payable is a contra account to Bonds Payable, a decrease in the company's liabilities. Subtracting the discount from Bonds Payable yields the carrying amount of the bonds. Thus, Jardine's liability is \$96,139, which is the amount the company effectively borrowed.

What Is the Interest Expense on These Bonds Payable?

Jardine pays interest on bonds semi-annually, which is common practice. Each semi-annual interest payment is set by the bond contract and therefore remains the same over the life of the bonds:

$$\begin{aligned}\text{Semi-annual interest payment} &= \$100,000 \times 0.09 \times 6/12 \\ &= \$4,500\end{aligned}$$

But Jardine's interest expense increases as the bonds march toward maturity. Remember: these bonds were issued at a discount.

Panel A of Exhibit 9-4 repeats the Jardine bond example we've been using. Panel B provides an amortization table that does three things:

1. determines the periodic interest expense (column C);
2. shows the amount of bond discount amortization (column E); and
3. shows the bond carrying amount (column G).

Study Exhibit 9-4 carefully because the amounts we'll be using come directly from the amortization table. This exhibit shows the effective-interest method of amortization, which is required by IAS 39 to measure interest expense.

Exhibit 9-4 Debt Amortization for a Bond Discount

Panel A – Bond Data

A1				
	A	B	C	D
1	Issue date	1 January 20X1	Maturity date	1 January 20X6
2	Face (or par or maturity) value	\$100,000	Number of periods	10 periods
3	Stated interest rate (per year)	9% per year	Market interest rate at issuance (per year)	10% per year
4	Stated interest rate (per period)	4.5% per period	Market interest rate at issuance (per period)	5% per period
5	Issue price	\$96,139	=	PV = (market rate, number of periods, interest payment, face value)

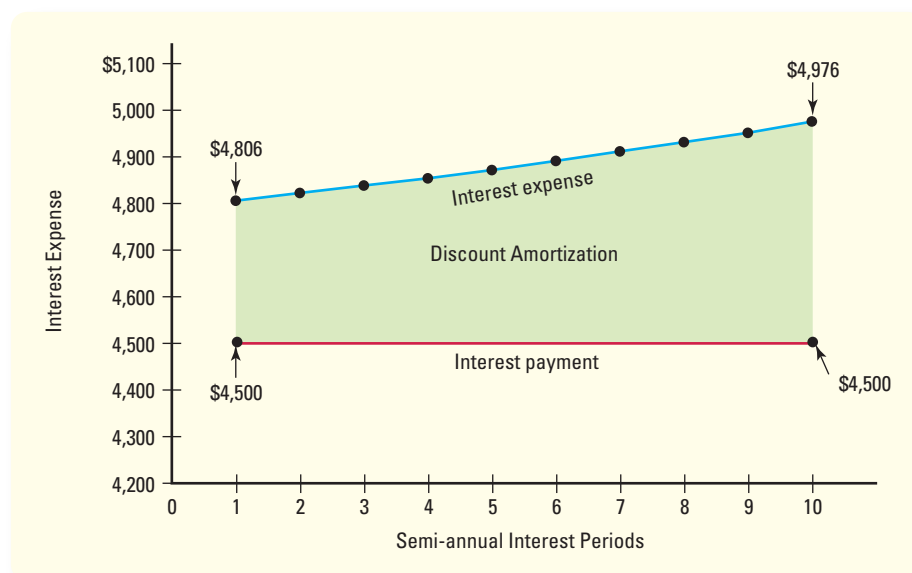
Panel B – Bond Amortization Table

A1							
	A	B	C	D	E	F	G
1	Semi-annual Interest Payment Dates	Beginning Bond Carrying Amount	Interest Expense (Market Rate × B)	Interest Payment (Bond Rate × Face Value)	Bond Discount Amortization (C – D)	Bond Discount Account Balance (Preceding F – E)	Ending Bond Carrying Amount (Face Value – F)
2	(Issuance) 1 Jan 20X1					\$ 3,861	\$ 96,139
3	(#1) 1 Jul 20X1	\$ 96,139	\$ 4,807	\$ 4,500	\$ 307	3,554	96,446
4	(#2) 1 Jan 20X2	96,446	4,822	4,500	322	3,232	96,768
5	(#3) 1 Jul 20X2	96,768	4,838	4,500	338	2,893	97,107
6	(#4) 1 Jan 20X3	97,107	4,855	4,500	355	2,538	97,462
7	(#5) 1 Jul 20X3	97,462	4,873	4,500	373	2,165	97,835
8	(#6) 1 Jan 20X4	97,835	4,892	4,500	392	1,773	98,227
9	(#7) 1 Jul 20X4	98,227	4,911	4,500	411	1,362	98,638
10	(#8) 1 Jan 20X5	98,638	4,932	4,500	432	930	99,070
11	(#9) 1 Jul 20X5	99,070	4,954	4,500	454	476	99,524
12	(#10) 1 Jan 20X6	99,524	4,976	4,500	476	0	100,000
13							
14	Notes:						
15	Column B	Beginning bond carrying amount, which is equal to preceeding Column G					
16		For issuance, carrying amount is the face value of the bond less bond discount (Column F)					
17	Column C	Interest expense each period, calculated as market rate per period × the beginning bond carrying amount in Column B					
18	Column D	Fixed interest payment each period, calculated as bond rate per period × the face value of the bond					
19	Column E	Bond discount amortization is the difference between interest expense (Column C) and interest payment (Column D)					
20	Column F	Bond discount balance is the preceeding balance (Column F) adjusted for current period amortization (Column E)					
21	Column G	Ending bond carrying amount, which is face value of the bond less the bond discount account balance (Column F)					
22							

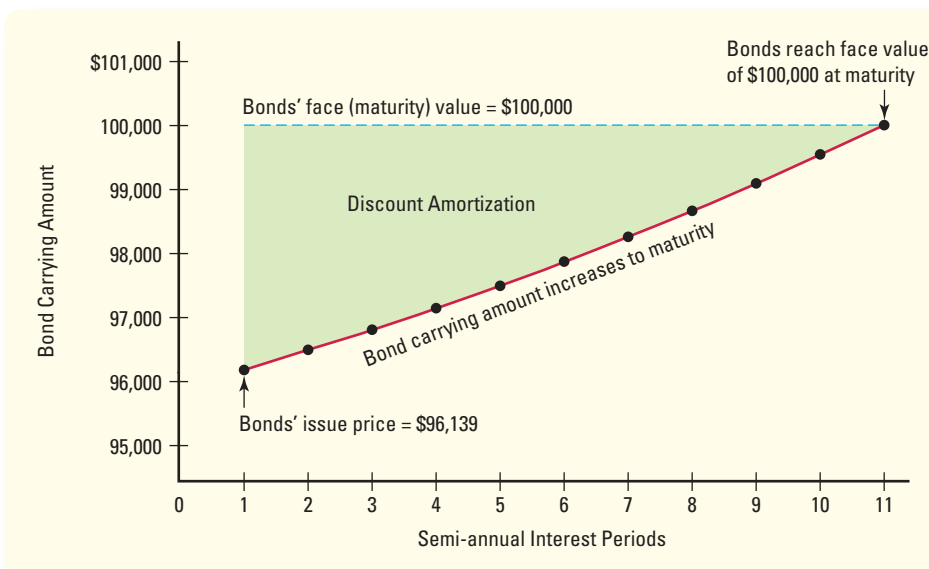
Interest Expense on Bonds Issued at a Discount

In Exhibit 9-4, Jardine borrowed \$96,139 cash but must pay \$100,000 when the bonds mature. What happens to the \$3,861 balance of the discount account over the life of the bond issue?

The \$3,861 is an additional interest expense to Jardine over and above the stated interest that Jardine pays each six months. Exhibit 9-5 graphs the interest expense and the interest payment on the Jardine bonds over their lifetime. Observe that the semi-annual interest payment is fixed—by contract—at \$4,500. But the amount of interest expense increases as the discount bond marches upward towards maturity. These amounts come from Exhibit 9-4 Panel B column C.

Exhibit 9-5 Interest Expense on Bonds Payable Issued at a Discount

The discount is allocated to interest expense through amortization over the term of the bonds. Exhibit 9-6 illustrates the amortization of the bonds from \$96,139 at the start to \$100,000 at maturity. These amounts come from Exhibit 9-4, column G.

Exhibit 9-6 Amortizing Bonds Payable Issued at a Discount

Now let's see how Jardine would account for these bonds issued at a discount. In our example, Jardine issued its bonds on January 1, 20X1. On July 1, Jardine made the first semi-annual interest payment. But Jardine's interest expense is greater than its payment of \$4,500. Jardine's journal entry to record interest expense and the interest payment for the first 6 months is as follows (with all amounts taken from Exhibit 9-4):

A1				
	A	B	C	D
1	20X1			
2	Jul 1	Interest Expense	4,807	
3		Discount on Bonds Payable		307
4		Cash		4,500
5		<i>To pay semi-annual interest and amortize bond discount.</i>		
6				

The credit to Discount on Bonds Payable accomplishes two purposes:

1. It adjusts the carrying value of the bonds as they march upward towards maturity value.
2. It amortizes the discount to interest expense.

At December 31, 20X1, Jardine accrues interest and amortizes the bonds for July through December with this entry (amounts from Exhibit 9-4):

A1				
	A	B	C	D
1	20X1			
2	Dec 31	Interest Expense	4,822	
3		Discount on Bonds Payable		322
4		Interest Payable		4,500
5		<i>To accrue semi-annual interest and amortize bond discount.</i>		
6				

At December 31, 20X1, Jardine's bond accounts appear as follows:

Bonds Payable		Discount on Bonds Payable	
	100,000	3,861	307
			322
		Bal. 3,232	

Bond carrying amount, \$96,768 = \$100,000 - \$3,232 from Exhibit 9-4, Panel B, row 4.

» Stop & Think

What would Jardine's 20X1 Income Statement and year-end Balance Sheet report for these bonds?

Answer:

Income Statement for 20X1

Interest expense (\$4,807 + \$4,822)	\$ 9,629
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Balance Sheet at December 31, 20X1

Current liabilities:	
Interest payable	\$ 4,500
Long-term liabilities:	
Bonds payable	\$100,000
Less: Discount on bonds payable	(3,232)
	96,768

At maturity on January 1, 20X6, the discount will have been amortized to zero, and the bonds' carrying amount will be the face value of \$100,000. Jardine will retire the bonds by paying \$100,000 to the bondholders.

A1				
	A	B	C	D
1	20X6			
2	Jan 1	Bond Payable	100,000	
3		Cash		100,000
4		<i>To pay bond payable.</i>		
5				

Issuing Bonds Payable at a Premium

Let's modify the Jardine bond example to illustrate issuance of the bonds at a premium. Assume that Jardine issues \$100,000 of five-year, 9% bonds that pay interest semi-annually. If the 9% bonds are issued when the market interest rate is 8%, their issue price is \$104,055². The premium on these bonds is \$4,055, and Exhibit 9-7 shows how to amortize the bonds by the effective-interest method. In practice, bond premiums are rare because few companies issue their bonds to pay cash interest above the market interest rate. We cover bond premiums for completeness.

Jardine's entries to record issuance of the bonds on January 1, 20X1, is as follows:

A1				
	A	B	C	D
1	20X1			
2	Jan 1	Cash	104,055	
3		Bonds Payable		100,000
4		Premium on Bonds Payable		4,055
5		<i>To issue bonds at a premium.</i>		
6				

At the beginning, Jardine's liability is \$104,055—not \$100,000. The accounting equation makes this clear.

Assets	=	Liabilities	+	Shareholders' Equity
+ 104,055	=	+ 100,000	+	0
		+ 4,055		

Immediately after issuing the bonds at a premium on January 1, 20X1, Jardine would report the bonds payable on the Balance Sheet as follows:

Non-current liabilities:		
Bonds payable	\$100,000	
Premium on bonds payable	4,055	104,055

²Appendix B (available on www.pearsonglobaleditions.com/Harrison) shows how to determine the price of this bond. If you are using Excel, you can use the PV function, i.e., PV (market rate per period, number of periods, interest payment per period, and face value) or “=PV(4%, 10, -4500, -100000)” to get \$104,055. Any difference from workings using Appendix B is due to rounding of time value factors.

Exhibit 9-7 is similar in structure to Exhibit 9-4, which you saw earlier. The only difference is in the final column, where the balance of the bond premium account is added to the face value of the bonds to determine the carrying amount.

Exhibit 9-7 Debt Amortization for a Bond Premium

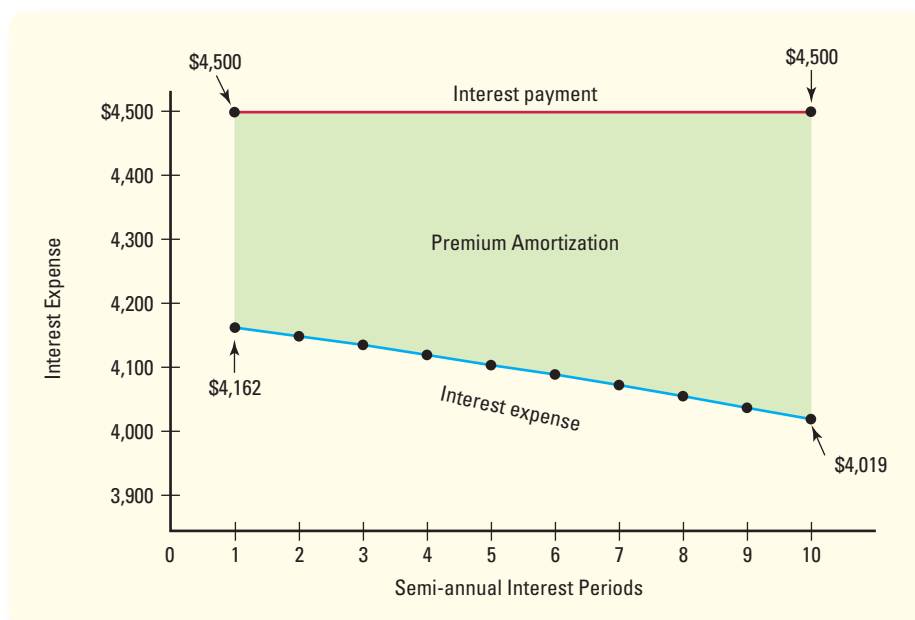
Panel A – Bond Data

A1				
	A	B	C	D
1	Issue date	1 January 20X1		Maturity date
2	Face (or par or maturity) value	\$100,000		Number of periods
3	Stated interest rate (per year)	9% per year		Market interest rate at issuance (per year)
4	Stated interest rate (per period)	4.5% per period		Market interest rate at issuance (per period)
5	Issue price	\$104,055	=	PV = (market rate, number of periods, interest payment, face value)

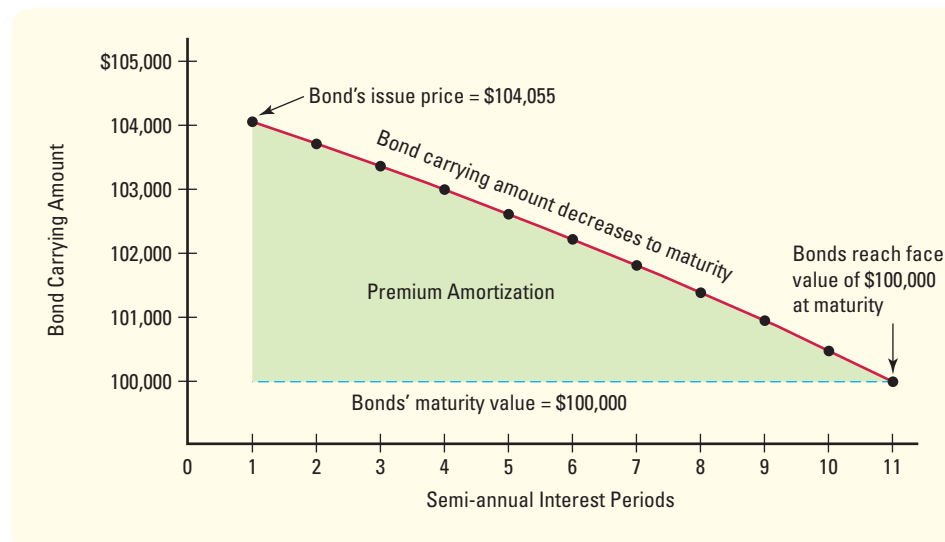
Panel B – Bond Amortization Table

A1							
	A	B	C	D	E	F	G
1	Semi-annual Interest Payment Dates	Beginning Bond Carrying Amount	Interest Expense (Market Rate × B)	Interest Payment (Bond Rate × Face Value)	Bond Premium Amortization (C – D)	Bond Premium Account Balance (Preceding F – E)	Ending Bond Carrying Amount (Face Value + F)
2	(Issuance) Jan 1 20X1					\$ 4,055	\$ 104,055
3	(#1) 1 Jul 20X1	\$ 104,055	\$ 4,162	\$ 4,500	\$ 338	3,718	103,718
4	(#2) 1 Jan 20X2	103,718	4,149	4,500	351	3,366	103,366
5	(#3) 1 Jul 20X2	103,366	4,135	4,500	365	3,001	103,001
6	(#4) 1 Jan 20X3	103,001	4,120	4,500	380	2,621	102,621
7	(#5) 1 Jul 20X3	102,621	4,105	4,500	395	2,226	102,226
8	(#6) 1 Jan 20X4	102,226	4,089	4,500	411	1,815	101,815
9	(#7) 1 Jul 20X4	101,815	4,073	4,500	427	1,388	101,388
10	(#8) 1 Jan 20X5	101,388	4,056	4,500	444	943	100,943
11	(#9) 1 Jul 20X5	100,943	4,038	4,500	462	481	100,481
12	(#10) 1 Jan 20X6	100,481	4,019	4,500	481	0	100,000
13							
14	Notes:						
15	Column B	Beginning bond carrying amount, which is equal to preceding Column G					
16		For issuance, carrying amount is the face value of the bond <i>plus</i> bond premium (Column F)					
17	Column C	Interest expense each period, calculated as market rate per period × the beginning bond carrying amount in Column B					
18	Column D	Fixed interest payment each period, calculated as bond rate per period × the face value of the bond					
19	Column E	Bond premium amortization is the difference between interest expense (Column C) and interest payment (Column D)					
20	Column F	Bond premium balance is the preceding balance (Column F) adjusted for current period amortization (Column E)					
21	Column G	Ending bond carrying amount, which is face value of the bond <i>plus</i> the bond premium account balance (Column F)					
22							

In Exhibit 9-7, Jardine borrowed \$104,055 cash but must pay back only \$100,000 at maturity. The \$4,055 premium is a reduction in Jardine's interest expense over the term of the bonds. Exhibit 9-8 graphs Jardine's interest payments (column D from Exhibit 9-7) and interest expense (column C).

Exhibit 9-8 Interest Expense on Bonds Payable Issued at a Premium

Amortization of the bond's premium decreases interest expense each period over the term of the bonds. Exhibit 9-9 illustrates the amortization of the bonds from the issue price of \$104,055 to a maturity value of \$100,000. All amounts are taken from Exhibit 9-7.

Exhibit 9-9 Amortizing Bonds Payable Issued at a Premium

When Jardine makes the first interest payment on July 1, 20X1, the following entry will be recorded, based on row 3 of panel B, Exhibit 9-7.

A1				
	A	B	C	D
1	20X1			
2	Jul 1	Interest Expense	4,162	
3		Premium on Bonds Payable	338	
4		Cash		4,500
5		<i>To pay semi-annual interest and amortize bond premium.</i>		
6				

Similarly, when the bond is retired on January 1, 20X6, the premium account will have been amortized down to zero balance and the only thing to record will be the payment of the bond principal to bond holders.

A1				
	A	B	C	D
1	20X6			
2	Jan 1	Bond Payable	100,000	
3		Cash		100,000
4		<i>To pay bond payable.</i>		
5				

Partial-Period Interest Amounts

Companies don't always issue bonds at the beginning or the end of their accounting year. They issue bonds when market conditions are most favorable, and that may be on May 16, August 1, or any other date. To illustrate partial-period interest, assume Fuji-Xerox issues \$100,000 of 8% bonds payable at 96 on August 31, 20X0. The market rate of interest was 9%, and these bonds pay semi-annual interest on February 28 and August 31 each year. The first few lines of Fuji-Xerox's amortization table are:

Semi-annual Interest Date	4% Interest Payment	4½% Interest Expense	Discount Amortization	Discount Account Balance	Bond Carrying Amount
Aug 31, 20X1				\$4,000	\$96,000
Feb 28, 20X2	\$4,000	\$4,320	\$320	3,680	96,320
Aug 31, 20X2	4,000	4,334	334	3,346	96,654

Fuji-Xerox's accounting year ends on December 31, so at year-end Fuji-Xerox must accrue interest and amortize the bond discount for four months (September through December). At December 31, 20X0, Fuji-Xerox will make this entry:

A1				
	A	B	C	D
1	20X0			
2	Dec 31	Interest Expense ($\$4,320 \times 4/6$)	2,880	
3		Discount on Bonds Payable ($\$320 \times 4/6$)		213
4		Interest Payable ($\$4,000 \times 4/6$)		2,667
5		<i>To accrue interest and amortize discount at year-end.</i>		
6				

The year-end entry at December 31, 20X0, uses 4/6 of the upcoming semi-annual amounts at February 28, 20X1. This example clearly illustrates the benefit of an amortization schedule.

The Straight-line Amortization Method: A Quick Way to Measure Interest Expense

While not allowed by *IAS 39*, there's a less precise way to amortize bond discount or premium. The straight-line amortization method divides a bond discount (or premium) into equal periodic amounts over the bond's term. The amount of interest expense is the same for each interest period. This method is simply a "quick and dirty" way to estimate interest expense when you do not have the tools to use the effective interest rate method as required by *IAS 39*. Unless otherwise stated, you should always use the effective interest method as previously discussed.

Let's apply the effective interest method to the Jardine bonds issued at a discount and illustrated in Exhibit 9-4. Suppose Jardine's Chief Financial Officer is considering issuing the 9% bonds at \$96,139. To quickly estimate semi-annual interest expense on the bonds, she can use the straight-line interest amortization method for the bond discount, as an approximation, as follows:

Semi-annual cash interest payment ($\$100,000 \times 0.09 \times 6/12$)	\$4,500
+ Semi-annual amortization of discount ($\$3,861 \div 10$)	386
= Estimated semi-annual interest expense	<u>\$4,886</u>

The straight-line interest amortization method uses this same amortization amount for every period over the term of the bonds.

Jardine's entry to record interest and amortization of the bond discount under the straight-line interest amortization method would be:

A1				
	A	B	C	D
1	20X1			
2	Jul 1	Interest Expense	4,886	
3		Discount on Bonds Payable		386
4		Cash		4,500
5		<i>To pay semi-annual interest and amortize bond discount.</i>		
6				

Should We Retire Bonds Payable Before Their Maturity?

Normally, companies wait until maturity to pay off, or retire, their bonds payable. But companies sometimes retire bonds early. The main reason for retiring bonds early is to relieve the pressure of making high interest payments. Also, the company may be able to borrow at a lower interest rate.

Some bonds are **callable**, which means that the issuer may call, or pay off, those bonds at a prearranged price (this is the call price) whenever the issuer chooses. The call price is often a percentage point or two above the par value, perhaps 101 or 102. Callable bonds give the issuer the benefit of being able to pay off the bonds whenever it is most favorable to do so. The alternative to calling the bonds is to purchase them in the open market at their current market price.

Assume that Jardine has \$300 million of debenture bonds outstanding and the unamortized discount is \$30 million. The bonds are callable at the discretion of the management at 101. Lower interest rates may convince management to pay off these bonds now. If the market price of the bonds is 99, will Jardine call the bonds at 101 or purchase them for 99 in the open market? Market price is the better choice because the market price is lower than the call price. Let's see how to account for an early retirement of bonds payable. Retiring the bonds at 99 results in a loss of \$27 million, computed as follows:

	Millions
Par value of bonds being retired.....	\$300
Less: Unamortized discount.....	(30)
Carrying amount of the bonds being retired.....	270
Market price ($\$300 \times 0.99$).....	297
Loss on retirement of bonds payable.....	<u>\$ 27</u>

Gains and losses on early retirement of bonds payable are usually reported as Other income (loss) on the Income Statement.

Convertible Bonds and Notes

Some corporate bonds may be converted into the issuing company's share capital. These bonds are called **convertible bonds** (or convertible notes). For investors, these bonds combine the safety of (a) assured receipt of interest and principal on the bonds with (b) the opportunity for gains on the shares. The conversion feature can be so attractive that investors are willing to accept a lower interest rate than they would on non-convertible bonds. The lower cash interest payments benefit the issuer. If the market price of the issuing company's shares gets high enough, the bondholders will convert the bonds into shares.

Suppose Jardine has convertible notes payable of \$100 million. If Jardine's share price rises high enough, the noteholders will convert the notes into the company's shares. Conversion of the notes payable into shares will decrease Jardine's liabilities and increase its equity.

Assume the noteholders convert the notes into 4 million shares of Jardine shares (\$1 par) on May 14. Jardine makes the following entry in its accounting records:

A1				
	A	B	C	D
1	May 14	Notes Payable	100,000,000	
2		Share Capital (4,000,000 × \$1 par)		4,000,000
3		Capital in Excess of Par		96,000,000
4		<i>To record conversion of notes payable.</i>		
5				

The accounting equation shows that liabilities decrease and equity goes up.

Assets	=	Liabilities	+	Shareholders' Equity
0	=	- 100,000,000	+	+ 4,000,000 + 96,000,000

The carrying amount of the notes (\$100 million) ceases to be debt and becomes shareholders' equity. Share Capital is recorded at its par value, which is a dollar amount assigned to each share. In this case, the credit to Share Capital is \$4,000,000 (4,000,000 shares × \$1 par value per share). The extra carrying amount of the notes payable (\$96,000,000) is credited to another shareholders' equity account, Capital in Excess of Par. We'll be discussing equity transactions in the next chapter.

ACCOUNT FOR LEASES

4 Account for leases

A **lease** is a rental agreement in which the tenant (**lessee**) agrees to make rent payments to the property owner (**lessor**) in exchange for the use of the asset. Leasing allows the lessee to acquire the use of a needed asset without having to make the large up-front payment that purchase agreements require.

Let's see how Jardine accounts for its leases.

ADAPTED EXCERPTS FROM JARDINE MATHESON'S NOTES TO THE ACCOUNTS

Leases

Leases are classified as *finance leases* when the terms of the lease transfer substantially all the risks and rewards of ownership to the lessee. All other leases are classified as *operating leases*.

Plant and machinery under *finance leases* are capitalized at the commencement of the lease at the lower of the fair value of the leased asset and the present value of the minimum lease payments. Lease payments are allocated between the liability and finance charges so as to achieve a constant rate on the finance balance outstanding.

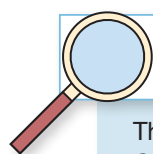
Payments made under *operating leases* are charged to profit and loss on a straight-line basis over the period of the lease.

Source: Jardine Matheson's 2015 annual report, page 44

Types of Leases

IAS 17—Leases provides for two categories of leases: operating lease and finance lease (sometimes referred to as financing or capital lease). The accounting treatments for the two types of leases are vastly different. It is therefore important to understand the criteria that would make a lease agreement classified as an operating lease or a capital lease. *IAS 17* states that if the terms of the lease meet *any* of the following conditions, it will be recognized as a capital lease:

- The lease transfers substantially all risks and rewards of the asset to the lessee.
- The lease transfers ownership of the asset to the lessee at the end of the lease.
- The lease term represents a substantial part of the asset's useful life.
- The present value of the lease payments represents a substantial part of the fair value of the asset.



A Closer Look

This is a good example of how the overall approaches of IFRSs differ from U.S. GAAPs. Rather than specific rules or “bright lines,” IFRSs employs a principle-based approach that focuses on the overall substance of the transaction, rather than on the mechanical form, and that leaves more to the judgment of the preparer of the financial statement. For example, the recognition of a lease as a capital lease under U.S. GAAP would be that the term of the lease exceeds 75% of the asset's useful life, and the present value of the lease payments exceeds 90% of the asset's fair value, as opposed to the use of the “substantial part” principle in IFRS.

The FASB and IASB have recently issued a new replacement standard for Leases, *IFRS 16 – Leases*, which will be effective for financial statements beginning January 1, 2019. The new standard is a very significant departure from what we are using now, *IAS 17 – Leases*. It will substantially change the way we account for leases. The new standard focuses on “rights of use” as the key decision factor in determining the appropriate accounting treatment for leases. This means doing away with the capital versus operating lease classifications. As long as an entity has a rights of use of an asset with an accompanying commitment to make payments, they will be recognized on the books as assets (representing its rights to use the underlying leased asset) and liabilities (representing its obligation to make lease payments), respectively. There will be some “practical expedients” for very short term leases (or leases of items of very low value), which can continue to be paid as an expense.

Operating Leases. These are basically rental agreements between the lessor and lessee. Many operating leases are non-cancellable (or cancellable with substantial penalties), and require the lessee to commit funds to pay the lessor for use of the leased asset for years. They give the lessee

the right to use the asset but provide no continuing rights to the asset. Instead, the lessor retains the usual risks and rewards of owning the leased asset. To account for an operating lease, the lessee debits Rent Expense (or Lease Expense) and credits Cash for the amount of the lease payment.

Jardine incurred a total of \$1,085 million operating lease expense in the 2015 financial year ended December 31, 2011. Jardine would have performed a journal entry similar to the entry below:

A1				
	A	B	C	D
1	31 Dec	Lease expense	1,085 million	
2		Cash		1,085 million
3		<i>To record operating lease payments</i>		
4				

Assets	=	Liabilities	+	Shareholders' Equity	
- 154 million	=	0	+	- 154 million	(lease expense)

The properties or equipment being leased (via operating lease) are not reported as Jardine's assets on its Balance Sheet. Similarly, Jardine's non-cancellable future payments for its operating leases are also not recorded as liabilities on its Balance Sheet. Nevertheless, *IAS 17* imposes an additional disclosure requirement on these operating lease commitments. Jardine's operating lease commitments are shown below.

ADAPTED EXCERPTS FROM JARDINE MATHESON'S NOTES TO THE ACCOUNTS

Operating Lease Commitments (USD millions)	2015	2014
■ Due within one year	\$862	\$863
■ Due between one and two years	611	605
■ Due between two and three years	376	383
■ Due between three and four years	202	218
■ Due between four and five years	147	146
■ Due beyond five years	610	730
	2,808	2,945

Source: Jardine Matheson's 2015 annual report, page 102

Capital Leases (also called **Finance Leases**). Accounting for a **capital lease** is much like accounting for the purchase of an asset. The lessee enters the asset into the lessee's long-term asset accounts and records a long-term lease liability at the beginning of the lease term. Thus, the lessee capitalizes the asset even though the lessee may never take legal title to the asset, because the lease agreement makes the lessee assume the risks and rewards of ownership of the assets and the associated obligations.

When a capital lease is signed, the lessee will record the present value of its lease payments on its books. The original entry is a debit to lease assets and a credit to lease liability. As the asset is being used, it is depreciated in accordance to the entity's usual depreciation policy. When lease payments are made, it is first made against lease interest expense, and the remaining balance reduces the outstanding lease payments.

Let's assume on July 1, 20X1, Jardine signed a capital lease that requires a payment of \$10,000 per year for the next five years. The present value of the lease payments is determined to be \$43,295 (using a 5% interest rate).³ On signing this lease agreement, Jardine would record the following:

A1				
	A	B	C	D
1	20X1			
2	July 1	Lease Asset	43,295	
3		Lease Liability		43,295
4		<i>To record capital lease (\$10,000 per year for 5 years @5%).</i>		
5				

When the first payment is made, on June 30, 20X2, the \$10,000 payment is first applied to the interest for the lease liability for the period, which is \$2,165 ($\$43,295 \times 5\%$), and the balance of \$7,835 ($\$10,000$ less $\$2,165$) reduces the lease liability.

A1				
	A	B	C	D
1	20X2			
2	June 30	Lease Liability	7,835	
3		Interest Expense	2,165	
4		Cash		10,000
5		<i>To record first capital lease payment.</i>		
6				

The balance of the lease liability is now \$43,295 less \$7,835, or \$35,460. In the subsequent period, less and less interest will be charged, and higher portions of the lease payments will go towards reducing the lease liability. This schedule is called a lease amortization schedule, shown as follows.

A1						
	A	B	C	D	E	F
1	Period	Beginning Lease Liability	Payment	Towards Interest	Towards Lease Liability	Ending Lease Liability
2	(#1) 30 Jun 20X2	\$ 43,295	\$ 10,000	\$ 2,165	\$ 7,835	\$ 35,460
3	(#1) 30 Jun 20X3	35,460	10,000	1,773	8,227	27,233
4	(#2) 30 Jun 20X4	27,233	10,000	1,362	8,638	18,594
5	(#3) 30 Jun 20X5	18,594	10,000	930	9,070	9,524
6	(#4) 30 Jun 20X6	9,524	10,000	476	9,524	0
7						
8	Notes:					
9	Column B	At signing of contract, calculated as present value of lease payments, subsequently from ending balance (Column F)				
10	Column C	Payment per period				
11	Column D	Interest expense for the lease, calculated as lease rate $\times$ beginning lease liability (Column B)				
12	Column E	Principal payment towards lease liability, difference between payment (Column C) and interest expense (Column D)				
13	Column F	Ending lease liability which is equal to beginning balance (Column B) less principal lease liability payment (Column E)				
14						

³Appendix B (available on www.pearsonglobaleditions.com/Harrison) shows the present value calculations for leases. If you are using Excel, you can use the PV function, i.e., PV (market rate per period, number of periods, interest payment per period) or “= PV (5%, 5, -10,000)” to get \$43,295. Any difference from workings using Appendix B is due to rounding of time value factors.

ADAPTED EXCERPTS FROM JARDINE MATHESON'S NOTES TO THE ACCOUNTS

Finance Lease Assets (USD millions)	2015	2014
■ Plant and machinery	\$41	\$64
■ Motor vehicles	45	3
	<u>\$86</u>	<u>\$67</u>
 Finance Lease Liabilities (USD millions)	 2015	 2014
■ Current finance lease liabilities	\$31	\$36
■ Non-current finance lease liabilities	65	48
	<u>\$96</u>	<u>\$84</u>

Source: Jardine Matheson's 2015 annual report, pages 70, 102



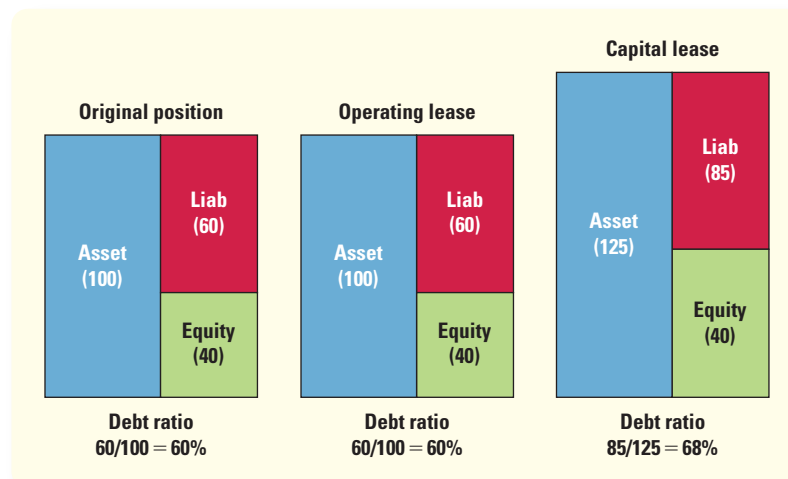
A Closer Look

While lease assets and lease liabilities start as the same amount, they don't always tally afterwards. Leased assets are usually depreciated using a straight-line basis (and may have residual values), while lease liability are amortized using an effective interest rate method. Furthermore, almost all lease arrangements will require payments at the beginning of the period rather than at the end of the period. You can refer to Appendix B (available on www.pearsonglobaleditions.com/Harrison) for further explanation.

Do Lessees Prefer Operating Leases or Capital Leases?

Suppose you were the chief financial officer (CFO) of a company contemplating a lease agreement. Your total assets are now \$100, funded by liability of \$60 and equity of \$40. The leases can be structured either as operating leases or as capital leases, with total lease payments of \$25. Which type of lease would you prefer? Why?

Let's use a general measure of indebtedness called *debt ratio* to illustrate the difference between operating and capital lease. We will discuss debt ratio at the end of the chapter, but for now, it is calculated as total liabilities divided by total assets. It shows how much of the operations are funded by non-shareholders. The basic impact of this lease is as follows:



Computing the debt ratio two ways (operating leases versus reclassifying them as capital leases) clearly shows that the addition of assets and liabilities when treated as capital lease will result in a higher debt ratio. You can see that a capital lease increases the debt ratio—by about eight percentage points in this example, but potentially by a lot more for other companies. By contrast, notice that operating leases

don't affect the debt ratio. For this reason, many companies prefer operating leases because they allow them to obtain the use of an asset without showing the obligations for the contractual payments.

Another way to think about it is that you will see most businesses having far more operating lease commitments than capital lease obligations. For example, you saw that Jardine has operating lease commitments of \$2,808 million and finance lease liabilities of just \$96 million. With the new *IFRS 16 – Leases*, most of these “commitments” will be recognized as “liabilities” from 2019 onwards.

ANALYZE THE ADVANTAGES AND DISADVANTAGES OF BORROWING

Financing Operations with Bonds or Shares?

Managers must decide how to get the money they need to pay for assets. There are three main ways to finance operations:

- by retained earnings
- by issuing shares
- by issuing bonds (or notes) payable

Each strategy has its advantages and disadvantages:

1. **Financed by retained earnings** means that the company already has enough cash to purchase the needed assets. There's no need to issue more shares or to borrow money. This strategy is low-risk to the company.
2. **Issuing shares** creates no liabilities or interest expense and is less risky to the issuing corporation. But issuing shares is more costly, as we shall see.
3. **Issuing bonds or notes payable** does not dilute control of the corporation. It often results in higher earnings per share because the earnings on borrowed money can potentially exceed interest expense. But creating more debt increases the risk of the company.

Earnings per share (EPS) is the amount of a company's net income for each of its shares outstanding. EPS is the single most important statistic for evaluating companies because EPS is a standard measure of operating performance that applies to companies of different sizes and from different industries. We will cover EPS in more detail in Chapter 10.

Suppose your business needs \$500,000 for expansion. Assume that it has net income of \$300,000 and 100,000 shares outstanding. You are considering two financing plans. Plan 1 is to issue \$500,000 of 6% bonds payable, and plan 2 is to issue 50,000 shares for \$500,000. You believe the new cash can be invested in operations to earn income of \$200,000 before interest and taxes.

Exhibit 9-10 shows the earnings-per-share advantage of borrowing. As you can see, your EPS amount is higher if you borrow by issuing bonds (compare lines 10 and 11). You will earn

5 Analyze the advantages and disadvantages of borrowing

Exhibit 9-10 Earnings-Per-Share Advantage of Borrowing

A1						
	A	B	C	D	E	F
				Plan 1		Plan 2
				Borrow \$500,000 at 6%		Issue 50,000 Shares of Share Capital for \$500,000
1						
2	Net income before expansion				\$ 300,000	\$ 300,000
3	Expected project income before interest and income tax			\$ 200,000		\$200,000
4	Less interest expense (\$500,000 × 0.06)			(30,000)		0
5	Expected project income before income tax			170,000		200,000
6	Less income tax expense (40%)			(68,000)		(80,000)
7	Expected project net income				102,000	120,000
8	Total company net income				\$ 402,000	\$ 420,000
9	Earnings per share after expansion:					
10	Plan 1 Borrow (\$402,000/100,000 shares)				\$ 4.02	
11	Plan 2 Issue Shares (\$420,000/150,000 shares)					\$2.80
12						



DECISION GUIDELINES

FINANCING WITH DEBT OR WITH SHARES

Nando’s is the leading chain of Portuguese/Mozambique-themed restaurants, famous for its flame-grilled chicken and “Peri-Peri” sauces. Originated in South Africa, Nando’s is now present in over 26 countries and has continued its expansion plan. Suppose Nando’s is expanding into a new country or region. Take the role of Fernando Duarte and Robert Brozin, founders of Nando’s, and assume you must make some key decisions about how to finance the expansion.



Decision	Guidelines
How will you finance Nando’s expansion?	Your financing plan depends on Nando’s ability to generate cash flow, your willingness to give up some control of the business, the amount of financing risk you are willing to take, and Nando’s credit rating. If yes, the business needs little outside financing. There is no need to borrow. If no, the business will need to issue additional shares or borrow the money.
Do Nando’s operations generate enough cash to meet all its financing needs?	If yes, then issue shares to other shareholders, who can vote their shares to elect the company’s directors. If no, then borrow from bondholders, or a financial institution, who have no vote in the management of the company.
Are you willing to give up some of your control of the business?	If much, then borrow as much as you can, and you may increase Nando’s earnings per share. But this will increase the business’s debt ratio and the risk of being unable to pay its debts. If little, then borrow sparingly. This will hold the debt ratio down and reduce the risk of default on borrowing agreements. But Nando’s earnings per share may be lower than if you were to borrow.
How much financing risk are you willing to take?	The better the credit rating, the easier it is to borrow on favorable terms. A good credit rating also makes it easier to issue shares. Neither shareholders nor creditors will entrust their money to a company with a bad credit rating.
How good is the business’s credit rating?	

more on the investment (\$102,000) than the interest it pays on the bonds (\$30,000). This is called **trading on the equity**, or using **leverage**. It is widely used to increase earnings per share.

In this case borrowing results in higher earnings per share than issuing new shares. Borrowing has its disadvantages, however. Interest expense may be high enough to eliminate net income and lead to losses. Also, borrowing creates liabilities that must be paid during bad years as well as good years. In contrast, a company that issues shares can omit its dividends during a bad year. The Decision Guidelines provide some help in deciding how to finance operations.

EVALUATE A COMPANY’S DEBT-PAYING ABILITY

Debt Ratio

6 Evaluate a company’s debt-paying ability

We have just seen how borrowing can increase EPS. But too much debt can lead to bankruptcy if the business cannot pay liabilities as they come due. Many companies in the recent financial crisis fell into the debt trap because they leveraged (borrowed) too much, in anticipation of an unending booming economy, which ran into a crisis eventually.

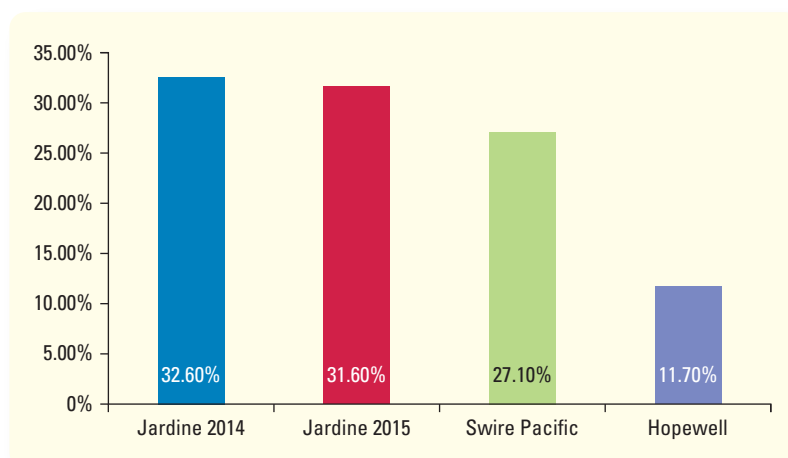
The **debt ratio** is a general measure of an entity's indebtedness. A simple version of the ratio expresses this indebtedness as the percentage of the assets that are financed by creditors. Jardine's 2015 debt ratio, for example, is its total liabilities (\$21,174, line 16 of its Balance Sheet, see start of chapter) divided by its total assets (\$66,955, line 4), which is 31.6%. It has decreased slightly from 32.6% (\$21,652 over \$66,457) in 2014.

Debt Ratio	Jardine 2015	Jardine 2014
$\frac{\text{Total liabilities}}{\text{Total assets}}$	$\frac{21,174}{66,955} = 31.6\%$	$\frac{21,652}{66,457} = 32.6\%$

It is difficult to find a company like Jardine. Its business interests span many different sectors and can only be described as a member of a “diversified industry.” It may sound counterlogical, but the best comparison to a diversified company is another diversified company even if their businesses are not exactly alike. They all have the power to enter and exit any industry if they wish. Let's use two other Hong Kong-based diversified companies: Swire Pacific (property, aviation, beverages, marine services, and trading and industrial) and Hopewell Holdings (property, expressway, power, and hotel and hospitality) for comparisons to Jardine Matheson (as shown in Exhibit 9-11).



Exhibit 9-11 Debt Ratio Comparisons

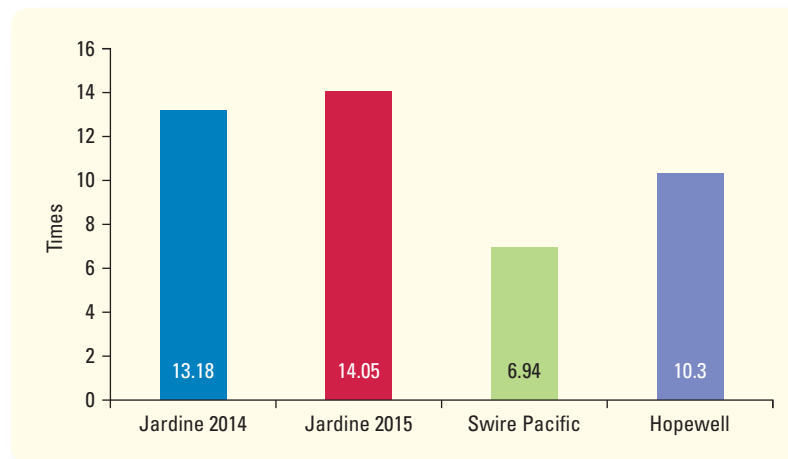


It looks like Jardine is more leveraged, i.e., it has higher debt ratio than the other two companies. However, in general, a debt ratio below 50% is considered low. Different businesses borrow different amounts based on their current needs, expansion plans, and financing model.

The Times-Interest-Earned Ratio

The debt ratio measures the effect of debt on the company's financial position but says nothing about the ability to pay interest expense. Analysts use a second ratio—the **times-interest-earned ratio**—to relate income to interest expense. To compute this ratio, we divide income from operations (also called operating income or operating profit) by interest expense. This ratio measures the number of times that operating income can cover interest expense. The times-interest-earned ratio is also called the **interest-coverage ratio**. A high times-interest-earned ratio indicates ease in paying interest expense; a low value suggests difficulty. As shown in Exhibit 9-12, Jardine's times-interest-earned ratio is at a healthy 13.18 times in 2014 and 14.05 in 2015. These ratios compare favorably to Swire Pacific (at 6.94 times) and Hopewell (at 10.30 times).

Times Interest Earned	Jardine 2015	Jardine 2014
$\frac{\text{Operating income}}{\text{Interest expense}}$	$\frac{3,779}{269} = 14.05 \text{ times}$	$\frac{3,676}{279} = 13.18 \text{ times}$

Exhibit 9-12 Times-Interest-Earned Ratio Comparisons

You may be wondering why Jardine's debt ratio is much higher than Swire Pacific and Hopewell, yet its times-interest-earned ratio is much better than the other two. Well, this is why you need to examine more than one financial ratio in order to obtain a more complete picture of the financial position and performance of a company. Much of Jardine's liabilities are non-interest bearing, so they do not incur borrowing costs. These include liabilities such as accounts payable, unearned revenue, provisions, and others. Some analysts prefer to calculate debt ratio purely based on interest-bearing borrowings instead of total liabilities.

Reporting Financing Activities on the Statement of Cash Flows

CASH FLOW

We started this chapter by looking at Jardine's Balance Sheet. It has current liabilities totaling \$12,602 million and non-current liabilities totaling \$8,572 million. Movements of cash related to the capital structure of a company, i.e., composition of long-term debt and equity, are reported in the Cash Flows from Financing Activities section of the Cash Flow Statement. We will discuss the Cash Flow Statement fully in Chapter 11, but for now, let's see how these items are reported by Jardine as illustrated in Exhibit 9-13.

Exhibit 9-13 Jardine Matheson's Consolidated Statement of Cash Flows

A1			
	A	B	C
	Jardine Matheson Holding Ltd. Consolidated Cash Flow Statement For the Year Ended December 31		
1			
2	(Adapted, in millions of US\$)	2015	2014
3	Cash flow from operating activities	\$ 4,118	\$ 3,354
4	Cash flow from investing activities	(3,229)	(2,303)
5	Cash flow from financing activities		
6	Issues of shares	2	2
7	Drawdown on borrowings	20,353	20,863
8	Repayment of borrowings	(20,337)	(20,576)
9	Other financing activities	(1,237)	(1,235)
10	Cash flows from financing activities	(1,219)	(946)
11	Net cash flow	\$ (330)	\$ (105)
12			

End-of-Chapter

Summary Problem

The Cessna Aircraft Company has an outstanding issue of 8% convertible bonds that mature in 20X8. Suppose the bonds are issued on October 1, 20X0, and pay interest each April 1 and October 1.

Requirements

- Complete the following effective-interest amortization table through October 1, 20X2.

Bond Data

Maturity (face) value—\$100,000

Stated interest rate—8%

Interest paid—4% semi-annually, \$4,000 ($\$100,000 \times 0.08 \times 6/12$)

Market interest rate at the time of issue—9% annually, 4½% semi-annually

Issue price—93.5 (use this price instead of calculating manually)

Semi-annual Interest Date	Amortization Table				
	A	B	C	D	E
	Interest Payment (4% of Maturity Amount)	Interest Expense (4½% of Preceding Bond Carrying Amount)	Discount Amortization (B – A)	Discount Account Balance (Preceding D – C)	Bond Carrying Amount (\$100,000 – D)
10-1-X0					
4-1-X1					
10-1-X1					
4-1-X2					
10-1-X2					

- Using the amortization table, record the following transactions:
 - Issuance of the bonds on October 1, 20X0.
 - Accrual of interest and amortization of the bonds on December 31, 20X0.
 - Payment of interest and amortization of the bonds on April 1, 20X1.
 - Conversion of one-third of the bonds payable into no-par share on October 2, 20X2. For no-par share, transfer the bond carrying amount into the Share Capital account. There is no Additional Paid-in Capital account.
 - Retirement of two-thirds of the bonds payable on October 2, 20X2. Purchase price of the bonds was based on their call price of 102.

Answers

Requirement 1

	A	B	C	D	E
Semi-annual Interest Date	Interest Payment (4% of Maturity Amount)	Interest Expense (4½% of Preceding Bond Carrying Amount)	Discount Amortization (B – A)	Discount Account Balance (Preceding D – C)	Bond Carrying Amount (\$100,000 – D)
10-1-X0				\$6,500	\$93,500
4-1-X1	\$4,000	\$4,208	\$208	6,292	93,708
10-1-X1	4,000	4,217	217	6,075	93,925
4-1-X2	4,000	4,227	227	5,848	94,152
10-1-X2	4,000	4,237	237	5,611	94,389

Requirement 2

A1				
	A	B	C	E
1	a.	20X0		
2		Oct 1	Cash ($\$100,000 \times 0.935$)	93,500
3			Discount on Bonds Payable	6,500
4			Bonds Payable	100,000
5			<i>To issue bonds at a discount.</i>	
6				
7	b.	Dec 31	Interest Expense ($\$4,208 \times 3/6$)	2,104
8			Discount on Bonds Payable ($\$208 \times 3/6$)	104
9			Interest Payable ($\$4,000 \times 3/6$)	2,000
10			<i>To accrue interest and amortize the bonds.</i>	
11				
12	c.	20X1		
13		Apr 1	Interest Expense ($\$4,208 \times 3/6$)	2,104
14			Interest Payable	2,000
15			Discount on Bonds Payable ($\$208 \times 3/6$)	104
16			Cash	4,000
17			<i>To pay semi-annual interest, part of which was accrued, and amortize the bonds.</i>	
18				
19				
20	d.	20X2		
21		Oct 2	Bonds Payable ($\$100,000 \times 1/3$)	33,333
22			Discount on Bonds Payable ($\$5,611 \times 1/3$)	1,870
23			Share Capital ($\$94,389 \times 1/3$)	31,463
24			<i>To record conversion of bonds payable.</i>	
25				
26	e.	Oct 2	Bonds Payable ($\$100,000 \times 2/3$)	66,667
27			Loss on Retirement of Bonds	5,074
28			Discount on Bonds Payable ($\$5,611 \times 2/3$)	3,741
29			Cash ($\$100,000 \times 2/3 \times 1.02$)	68,000
30			<i>To retire bonds payable before maturity.</i>	
31				

REVIEW | Liabilities

Quick Check (Answers are given at the end of the chapter.)

1. Which of the following is not an estimated liability?
 - a. Income taxes paid
 - b. Allowance for bad debts
 - c. Product warranties
 - d. Retirement obligations

2. The estimated warranty obligation at the end of the financial year is best described as which of the following?
 - a. Contingent liability
 - b. Unrecognized liability
 - c. Uncertain liability
 - d. Constructive liability
 - e. Liability

3. Crank the Volume grants a 120-day warranty on all stereos. Historically, approximately 1% of all units sold prove to be defective, requiring an average repair bill of \$100. Sales in March are \$472,500 for 4,500 units. In March, \$3,900 of defective units are returned for replacement. What entry must Crank the Volume make at the end of March to record the warranty expense?
 - a. Debit Warranty Expense and credit Provision for Warranty Repairs, \$4,725.
 - b. Debit Warranty Expense and credit Provision for Warranty Repairs, \$3,900.
 - c. Debit Warranty Expense and credit Cash, \$4,725.
 - d. No entry is needed at March 31.

4. Excursion Camera Co. was organized to sell a single product that carries a 45-day warranty against defects. Engineering estimates indicate that 4% of the units sold will prove defective and require an average repair cost of \$25 per unit. During Expedition's first month of operations, total sales were 800 units; by the end of the month, 15 defective units had been repaired. The liability for product warranties at month-end should be

a. \$1,175.	d. \$800.
b. \$425.	e. none of these.
c. \$375.	

5. A contingent liability should be recorded in the accounts
 - a. if the amount is due in cash within one year.
 - b. if the amount can be reasonably estimated.
 - c. if the related future event will probably occur.
 - d. Both b and c
 - e. Both a and c

6. An unsecured bond is a

a. serial bond.	d. mortgage bond.
b. term bond.	e. debenture bond.
c. registered bond.	

7. The Discount on Bonds Payable account
 - a. is expensed at the bond's maturity.
 - b. is a contra account to Bonds Payable.
 - c. is an expense account.
 - d. is a miscellaneous revenue account.
 - e. has a normal credit balance.

8. The discount on a bond payable becomes
 - a. additional interest expense over the life of the bonds.
 - b. a liability in the year the bonds are sold.
 - c. a reduction in interest expense over the life of the bonds.
 - d. additional interest expense the in year the bonds are sold.
 - e. a reduction in interest expense in the year the bonds mature.
9. A bond that matures in installments is called a
 - a. secured bond.
 - b. term bond.
 - c. serial bond.
 - d. callable bond.
 - e. zero coupon.
10. The carrying value of Bonds Payable equals
 - a. Bonds Payable + Discount on Bonds Payable.
 - b. Bonds Payable – Premium on Bonds Payable.
 - c. Bonds Payable – Discount on Bonds Payable.
 - d. Bonds Payable + Accrued Interest.
11. A corporation issues bonds that pay interest each May 1 and November 1. The corporation's December 31 adjusting entry may include a
 - a. credit to Cash.
 - b. debit to Interest Payable.
 - c. debit to Cash.
 - d. credit to Discount on Bonds Payable.
 - e. credit to Interest Expense.

Use this information to answer questions 12–16.

McCabe Corporation issued \$560,000 of 7% 10-year bonds. The bonds are dated and sold on January 1, 20X1. Interest payment dates are January 1 and July 1. The bonds are issued for \$521,724 to yield the market interest rate of 8%. Use the effective-interest method for questions 12–16.

12. What is the amount of interest expense that McCabe Corporation will record on July 1, 20X1, the first semi-annual interest payment date? (All amounts rounded to the nearest dollar.)
 - a. \$22,400
 - b. \$39,200
 - c. \$19,600
 - d. \$20,869
13. What is the amount of discount amortization that McCabe Corporation will record on July 1, 20X1, the first semi-annual interest payment date?
 - a. \$0
 - b. \$1,269
 - c. \$2,240
 - d. \$2,538
14. What is the total cash payment for interest for each 12-month period? (All amounts rounded to the nearest dollar.)
 - a. \$22,400
 - b. \$41,789
 - c. \$39,200
 - d. \$44,800
15. What is the total interest expense for the year ended December 31, 20X1?
 - a. \$41,789
 - b. \$41,879
 - c. \$39,200
 - d. \$19,600
16. What is the carrying amount of the bonds on the January 1, 20X2, Balance Sheet?
 - a. \$524,313
 - b. \$521,724
 - c. \$522,993
 - d. \$550,000

Accounting Vocabulary

accrued expense (p. 505) An expense incurred but not yet paid in cash. Also called accrued liability.

accrued liability (p. 505) A liability for an expense that has not yet been paid. Also called accrued expense.

bonds payable (p. 516) Groups of notes payable issued to multiple lenders called bondholders.

callable bond (p. 529) Bonds that are paid off early at a specified price at the option of the issuer.

capital lease (p. 532) Lease agreement in which the lessee assumes, in substance, the risks and rewards of asset ownership.

convertible bonds (or notes) (p. 530) Bonds or notes that may be converted into the issuing company's share capital at the investor's option.

current portion of long-term debt (p. 512) The amount of the principal that is payable within one year.

debentures (p. 516) Unsecured bonds—bonds backed only by the good faith of the borrower.

debt ratio (p. 537). A measure of indebtedness or leverage. Measured by total liabilities divided by total assets.

deposits (p. 506) Sometimes required by businesses before providing goods or services. May be deposits placed (as a customer) or deposits received (as a business).

discount (on a bond) (p. 516) Excess of a bond's face (par) value over its issue price.

earnings per share (EPS) (p. 535) Amount of a company's net income per share of its outstanding share capital.

interest-coverage ratio (p. 537) Another name for the times-interest-earned ratio.

lease (p. 530) Rental agreement in which the tenant (lessee) agrees to make rent payments to the property owner (lessor) in exchange for the use of the asset.

lessee (p. 530) Tenant or recipient of the leased asset in a lease agreement.

lessor (p. 530) Property owner of the leased asset in a lease agreement.

leverage (p. 536) Using borrowed funds to increase the return on equity. Successful use of leverage means earning

more income on borrowed money than the related interest expense, thereby increasing the earnings for the owners of the business. Also called trading on the equity.

market interest rate (p. 518) Interest rate that investors demand for loaning their money. Also called effective interest rate.

operating lease (p. 532) A lease in which the lessee does not assume the risks or rewards of asset ownership.

payroll (p. 507) Employee compensation, a major expense of many businesses.

pension (p. 508) Employee compensation that will be paid during the employee's retirement.

premium (on a bond) (p. 516) Excess of a bond's issue price over its face (par) value.

present value (p. 517) Amount a person would invest now to receive a greater amount at a future date.

serial bonds (p. 516) Bonds that mature in installments over a period of time.

short-term notes payable (p. 511) Note payable due within one year.

stated interest rate (p. 518) Interest rate that determines the amount of cash interest the borrower pays and the investor receives each year.

term bonds (p. 516) Bonds that all mature at the same time for a particular issue.

times-interest-earned ratio (p. 537) Ratio of income from operations to interest expense. Measures the number of times that operating income can cover interest expense. Also called the interest-coverage ratio.

trading on the equity (p. 536) Earning more income on borrowed money than the related interest expense, thereby increasing the earnings for the owners of the business. Also called leverage.

underwriter (p. 516) Organization that purchases the bonds from an issuing company and resells them to its clients or sells the bonds for a commission, agreeing to buy all unsold bonds.

ASSESS YOUR PROGRESS

Short Exercises

S9-1. (Learning Objective 1: Accounting for a note payable) Ferdie Sports Authority purchased inventory costing \$4,000 by signing an 8% short-term note payable. The purchase occurred on September 30, 20X0. Ferdie pays annual interest each year on September 30. Journalize the company's (a) purchase of inventory, (b) accrual of interest expense on June 30, 20X1, which is the year-end, and (c) payment of the note plus interest on September 30, 20X1. (Round your answers to the nearest whole number.)

LO 1

- L0 1** **S9-2. (Learning Objective 1: Reporting a short-term note payable and the related interest in the financial statements)** This short exercise works with Short Exercise 9-1.
1. Refer to the data in Short Exercise 9-1. Show what the company would report on its Balance Sheet at June 30, 20X1, and on its Income Statement for the year ended on that date.
 2. What single item will the financial statements for the year ended June 30, 20X2, report? Identify the financial statement, the item, and its amount.
- L0 1** **S9-3. (Learning Objective 1: Accounting for warranty expense and provision for warranty repairs)** Lawry guarantees automobiles against defects for five years or 50,000 miles, whichever comes first. Assume that a Lawry dealer in Paris, France, made sales of \$485,000 during 20X0. Lawry expects warranty costs during the five-year period to add up to about \$29,100. Lawry received cash for 30% of the sales and took notes receivable for the remainder. Payments to satisfy customer warranty claims totaled \$18,000 during 20X0.
1. Record the sales, warranty expense, and warranty payments for Lawry.
 2. Post to the Provision for Warranty Repairs T-account. The beginning balance was \$11,000. At the end of 20X0, how much in provision for warranty repairs does Lawry owe to its customers?
- L0 1** **S9-4. (Learning Objective 1: Applying accounting standards; reporting warranties in the financial statements)** Refer to the data given in Short Exercise 9-3. What amount of warranty expense will Lawry report during 20X0? Which accounting principle addresses this situation? Does the warranty expense for the year equal the year's cash payments for warranties? Explain the relevant accounting principle as it applies to measuring warranty expense.
- L0 1** **S9-5. (Learning Objective 1: Interpreting a company's contingent liabilities)** OC Petroleum Inc., an oil company, included a disclosure in its annual report stating that it was denying a variety of litigations from environmental groups that arose following an incident on one of their rigs in the Gulf of Mexico.
- How can a contingency liability become a real liability for the company? What would be the impact on the financial statements if these claims are recognized as a provision instead of contingent liability?
- L0 2** **S9-6. (Learning Objective 2: Pricing bonds)** Compute the cash received from the issuance of the following bonds:
- a. \$400,000 issued at 75.75
 - b. \$400,000 issued at 102.75
 - c. \$400,000 issued at 94.50
 - d. \$400,000 issued at 104.50
- L0 2** **S9-7. (Learning Objective 2: Determining bond prices at par, discount, or premium)** Determine whether the following bonds payable will be issued at maturity value, at a premium, or at a discount:
- a. The market interest rate is 5%. Esme Corp. issues bonds payable with a stated rate of 4.5%.
 - b. Trinity Corp. issued 5% bonds when the market interest rate was 5%.
 - c. Ottawa Company issued bonds payable that pay stated interest of 6%. At issuance, the market interest rate was 7.25%.
 - d. Howler, Inc., issued 7% bonds payable when the market rate was 6.75%.
- L0 2** **S9-8. (Learning Objective 2: Journalizing basic bond payable transactions; bonds issued at par)** Derp Corp. issued 15-year bonds payable with a face amount of \$70,000, when the market interest rate was 6.5%. The bonds were issued at par. Assume that the accounting year of Derp ends on December 31. Journalize the following transactions for Derp. Include an explanation for each entry.
- a. Issuance of the bonds payable at par on July 1, 20X0.
 - b. Accrual of interest expense on December 31, 20X0 (rounded to the nearest dollar).
 - c. Payment of cash interest on January 1, 20X1.
 - d. Payment of the bonds payable at maturity.

S9-9. (Learning Objectives 2: Issuing bonds payable; amortizing bonds by the effective-interest method) GIT, Inc., issued \$60,000 of 5%, 12-year bonds payable on March 31, 20X0. The market interest rate at the date of issuance was 8%, and the GIT bonds pay interest semi-annually.

L0 2

1. Prepare an effective-interest amortization table for the bonds through the first three interest payments. Round amounts to the nearest dollar.
2. Record GIT, Inc.'s issuance of the bonds on March 31, 20X0, and payment of the first semi-annual interest amount and amortization of the bond discount on September 30, 20X0. Explanations are not required.

S9-10. (Learning Objectives 2, 3: Accounting for bonds payable; analyzing data on long-term debt) Use the amortization table that you prepared for GIT's bonds in Short Exercise 9-9 to answer the following questions:

L0 2 3

1. How much cash did GIT borrow on March 31, 20X0? How much cash will GIT pay back at maturity on March 31, 20Y2 (i.e., 12 years later).
2. How much cash interest will GIT pay each six months?
3. How much interest expense will GIT report on September 30, 20X0, and on March 31, 20X1? Why does the amount of interest expense increase each period?

S9-11. (Learning Objective 2: Issuing bonds payable; accruing interest; amortizing bonds by the effective interest method) Villa Drive-Ins Ltd. issued a \$520,000, 8%, 10-year bond payable on July 1, 20X0, when the market rate was 10%. Also assume that Villa's accounting year ends on December 31. Journalize the following transactions for Villa Drive-Ins Ltd., including an explanation for each entry:

L0 2

- a. Issuance of the bond payable on July 1, 20X0.
- b. Accrual of interest expense and amortization of bonds on December 31, 20X0. (Use the effective interest amortization method, and round amounts to the nearest dollar.)
- c. Payment of the first semi-annual interest amount on January 1, 20X1.

S9-12. (Learning Objective 3: Recording operating and capital lease) Lily Pan Enterprises entered into two lease agreements for 5 years on January 1, 20X0. Both leases require a payment of \$15,000 per year at the end of each year. Lease 1 is structured as an operating lease while Lease 2 is structured as a capital lease with an implicit interest rate of 5%. Prepare all relevant journal entries for both leases on:

L0 3

- a. The day of signing the lease contract
- b. The day of the first lease payment

S9-13. (Learning Objective 4: Computing earnings-per-share effects of financing with bonds versus shares) Speedtown Marina needs to raise \$3.5 million to expand the company. Speedtown Marina is considering the issuance of either:

L0 4

- \$3,500,000 of 8% bonds payable to borrow the money, or
- 100,000 shares of share capital at \$35 per share.

Before any new financing, Speedtown Marina expects to earn net income of \$300,000, and the company already has 100,000 shares of share capital outstanding. Speedtown Marina believes the expansion will increase income before interest and income tax by \$500,000. The income tax rate is 35%.

Prepare an analysis to determine which plan is likely to result in the higher earnings per share. Based solely on the earnings-per-share comparison, which financing plan would you recommend for Speedtown Marina?

LO 4

■ writing assignment

S9-14. (Learning Objective 4: Computing the times-interest-earned ratio) Kermit Plumbing Products Ltd. reported the following data in 20X0 (in billions):

	20X0
Net operating revenues.....	\$29.8
Operating expenses	24.6
Operating income.....	5.2
Non-operating items:	
Interest expense.....	(1.6)
Other	(0.2)
Net income.....	\$ 3.4

Compute Kermit's times-interest-earned ratio, and write a sentence to explain what the ratio value means. Would you be willing to lend Kermit \$1 billion? State your reason.

LO 5

S9-15. (Learning Objective 5: Reporting liabilities, including capital lease obligations)

Lovely Home, Inc., includes the following selected accounts in its general ledger at December 31, 20X0:

Bonds payable	\$402,000
Equipment.....	113,000
Current portion of bonds payable	50,000
Notes payable, long-term	320,000
Interest payable (due March 1, 20X1).....	2,000
Accounts payable	35,000
Discount on bonds payable (all long-term).....	11,000
Accounts receivable.....	28,000

Prepare the liabilities section of Lovely Home, Inc.'s Balance Sheet at December 31, 20X0, to show how the company would report these items. Report total current liabilities and total liabilities.

Exercises MyLab Accounting

Select A and B exercises can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select exercises using MyLab Accounting.

Group A

LO 1

E9-16A. (Learning Objective 1: Accounting for warranty expense and the related liability)

The accounting records of From the Athena Ceramics included the following balances at the end of the period:

Provision for Warranty Repairs	Warranty Expense
Beg. bal. 3,000	

From the Athena has determined that its warranty obligations at the end of the period are equal to \$12,000. During 20X0, the business paid \$54,000 to satisfy the warranty claims.

Requirements

1. Journalize From the Athena's warranty expense for the period and the company's cash payments to satisfy warranty claims. Explanations are not required.
2. Show what From the Athena will report on its Income Statement and Balance Sheet for this situation.
3. Which data item from Requirement 2 will affect From the Athena's current ratio? Will From the Athena's current ratio increase or decrease as a result of this item?

E9-17A. (Learning Objective 1: Recording and reporting current liabilities) TransWorld Publishing completed the following transactions for one subscriber during 20X0:

L0 1

Oct 1	Sold a one-year subscription, collecting cash of \$1,500, plus sales tax of 8%.
Nov 15	Remitted (paid) the sales tax to the TianJin Municipality.
Dec 31	Made the necessary adjustment at year-end.

Requirement

1. Journalize these transactions (explanations not required). Then report any liability on the company's Balance Sheet at December 31, 20X0.

E9-18A. (Learning Objective 1: Reporting payroll expense and liabilities) Star Talent Search has an annual payroll of \$220,000. In addition, the company incurs a payroll tax expense of 8%. At December 31, Star owes salaries of \$8,200 and payroll tax of \$700. The company will pay these amounts early next year. Show what Star will report for the foregoing on its Income Statement and year-end Balance Sheet.

L0 1

E9-19A. (Learning Objective 1: Recording note payable transactions) Assume that Gretel Company completed the following note-payable transactions.

L0 1

20X0	
May 1	Purchased delivery truck costing \$85,000 by issuing a one-year, 6% note payable.
Dec 31	Accrued interest on the note payable.
20X1	
May 1	Paid the note payable at maturity.

Requirements

1. How much interest expense must be accrued at December 31, 20X0? (Round your answer to the nearest whole dollar.)
2. Determine the amount of Gretel's final payment on May 1, 20X1.
3. How much interest expense will Gretel report for 20X0 and for 20X1? (Round your answer to the nearest whole dollar.)

E9-20A. (Learning Objective 1: Accounting for income tax) At December 31, 20X0, Sandara Real Estate reported a current liability for income tax payable of \$190,000. During 20X1, Sandara earned income of \$1,300,000 before income tax. The company's income tax rate during 20X1 was 36%. Also during 20X1, Sandara paid income taxes of \$360,000.

L0 1

How much income tax payable did Sandara Real Estate report on its Balance Sheet at December 31, 20X1? How much income tax expense did Sandara report on its 20X1 Income Statement?

E9-21A. (Learning Objectives 1, 5: Analyzing liabilities) Riverside Manors, Inc., builds environmentally sensitive structures. The company's 20X1 revenues totaled \$2,780 million, and at December 31, 20X1, the company had \$660 million in current assets. The December 31, 20X1 and 20X0, Balance Sheets reported the liabilities and shareholders' equity as follows:

L0 1 5

■ **writing assignment**

A1			
	A	B	C
1	At year-end (in millions)	20X1	20X0
2	Liabilities and Shareholders' Equity		
3	Current Liabilities		
4	Accounts payable	\$ 138	\$ 179
5	Accrued expenses	155	172
6	Employee compensation and benefits	38	20
7	Current portion of long-term debt	9	24
8	Total Current Liabilities	340	395
9	Long-Term Debt	1,494	1,323
10	Post-Retirement Benefits Payable	122	123
11	Other Liabilities	12	8
12	Shareholders' Equity	2,027	1,784
13	Total Liabilities and Shareholders' Equity	\$ 3,995	\$ 3,633
14			

Requirements

1. Describe each of Riverside Manors, Inc.'s liabilities and state how the liability arose.
2. What were the company's total assets at December 31, 20X1? Was the company's debt ratio at the end of 20X1 high, low, or in a middle range?

L0 1

E9-22A. (Learning Objective 1: Reporting a contingent liability) Rupert Security Systems' revenues for 20X0 totaled \$7.3 million. As with most companies, Rupert is a defendant in lawsuits related to its products. Note 14 of the Rupert Annual Report for 20X0 explains the company's policy on legal proceedings, stating that they will accrue for amounts related to legal matters if liability is probable and if it is reasonable to estimate an amount.

Requirements

1. Suppose Rupert's lawyers believe that a significant legal judgment against the company is reasonably possible. How should Rupert report this situation in its financial statements?
2. Suppose Rupert's lawyers believe it is probable that a \$1.8 million judgment will be rendered against the company. Report this situation in Rupert's financial statements. Journalize any entry requirements by IFRS. Explanations are not required.

L0 1 5

E9-23A. (Learning Objectives 1, 5: Reporting current and long-term liabilities) Assume that Boni Electronics completed these selected transactions during June 20X0:

- a. Sales of \$2,400,000 are subject to estimated warranty cost of \$168,000. The provision for warranty repairs at the beginning of the year was \$35,000, and warranty payments for the year totaled \$52,000.
- b. On June 1, Boni Electronics signed a \$56,000 note payable that requires annual payments of \$14,000 plus 6% interest on the unpaid balance each June 2.
- c. Music For You, Inc., a chain of music stores, ordered \$135,000 worth of CD players. With its order, Music For You, Inc., sent a check for \$135,000 in advance, and Boni shipped \$75,000 of the goods. Boni will ship the remainder of the goods on July 3, 20X0.
- d. The June payroll of \$270,000 is subject to employee withheld income tax of \$33,000 and payroll tax of 7.65%. On June 30, Boni pays employees their take-home pay and accrues all tax amounts.

Requirement

1. Report these items on Boni Electronics' Balance Sheet at June 30, 20X0.

L0 2

E9-24A. (Learning Objective 2: Issuing bonds payable [premium]; paying and accruing interest; amortizing the bonds by the effective interest method) On January 31, Driftwood Logistics, Inc., issued 10-year, 6% bonds payable with a face value of \$15,000,000. The bonds were issued when the market rate was 5% and pay interest on January 31 and July 31. Driftwood Logistics, Inc., amortizes bonds by the effective interest method.

Requirements

1. Record issuance of the bonds on January 31.
2. Record the semi-annual interest payment and amortization of bond discount on July 31.
3. Record the interest accrual and discount amortization on December 31.

E9-25A. (Learning Objectives: Measuring cash amounts for a bond payable [premium]; amortizing the bonds by the straight-line method) Federal Bank has \$600,000 of 7% debenture bonds outstanding. The bonds were issued at 103 in 20X0 and mature in 20Z0 (20 years later).

L0 2

Requirements

1. How much cash did Federal Bank receive when it issued these bonds?
2. How much cash in total will Federal Bank pay the bondholders through the maturity date of the bonds?
3. Take the difference between your answers to Requirements 1 and 2. What does this number represent?
4. Estimate the Federal Bank's interest expense using the straight-line amortization method for the first two interest payments.

E9-26A. (Learning Objective 2: Issuing bonds payable [discount]; recording interest payments and the related bond amortization) Goal Sports Ltd. is authorized to issue \$3,200,000 of 10%, 10-year bonds payable. On December 31, 20X0, when the market interest rate is 12%, the company issues \$2,500,000 of the bonds. Goal Sports Ltd. amortizes bond discounts by the effective-interest method. The semi-annual interest dates are June 30 and December 31.

L0 2

■ spreadsheet

Requirements

1. Prepare a bond amortization table for the first four semi-annual interest periods.
2. Record issuance of the bonds payable on December 31, 20X0, the first semi-annual interest payment on June 30, 20X1, and the second payment on December 31, 20X1.

E9-27A. (Learning Objective 2: Issuing bonds payable [premium]; recording interest accrual and payment and the related bond amortization) On June 30, 20X0, the market interest rate is 4%. Score Sports Ltd. issues \$850,000 of 5%, 30-year bonds payable. The bonds pay interest on June 30 and December 31. Score Sports Ltd. amortizes bonds by the effective-interest method.

L0 2

■ spreadsheet

Requirements

1. Prepare a bond amortization table for the first four semi-annual interest periods.
2. Record the issuance of bonds payable on June 30, 20X0, the payment of interest on December 31, 20X0, and the payment of interest on June 30, 20X1.

E9-28A. (Learning Objective 2: Creating a bond amortization schedule [discount]) Dracula Co. issued \$110,000 of 8%, 10-year bonds payable on January 1, 20X0, when the market interest rate was 10%. The company pays interest annually at year-end.

L0 2

■ spreadsheet

Requirement

1. Create a spreadsheet model to prepare a schedule to amortize the bonds. Use the effective-interest method of amortization. (Round to the nearest dollar.)

E9-29A. (Learning Objective 3: Recording operating and capital lease, preparing lease amortization schedule) Big Billy Guff entered into a lease agreement for an asset on the following terms:

L0 3

- Lease period = 4 years
- Lease payment = \$25,000 at the end of each period
- Lease interest rate = 10%

Requirements

1. Assuming that this lease qualifies as a capital lease, prepare the lease amortization table, starting from the signing of the lease to the final lease payments. Prepare the journal entry for the first lease payment.

- Assuming that this lease qualifies as an operating lease, prepare the journal entry for the first lease payment.
- Explain how a lease for the same item may be treated so differently by two companies.

L0 4

E9-30A. (Learning Objective 4: Measuring the times-interest-earned ratio) Companies that operate in different industries may have very different financial ratio values. These differences may grow even wider when we compare companies located in different countries.

A1				
	A	B	C	D
1	(Amounts in millions or billions)	Company A	Company N	Company S
2	Income data			
3	Total revenues	\$ 9,723	¥ 7,311	€ 136,431
4	Operating income	291	222	5,581
5	Interest expense	42	31	671
6	Net income	27	15	441
7				
8	Asset and liability data			
9	(Amounts in millions or billions)			
10	Total current assets	430	5,943	170,150
11	Long-term assets	138	48	45,324
12	Total current liabilities	196	2,198	72,420
13	Long-term liabilities	139	2,350	110,757
14	Shareholders' equity	233	1,443	32,297
15				

Compare three leading companies on their current ratio, debt ratio, and times-interest-earned ratio. Compute three ratios for Company A, Company N, and Company S.

Based on your computed ratio values, which company looks the least risky?

L0 4**■ writing assignment**

E9-31A. (Learning Objective 4: Analyzing alternative plans for raising money) First Bank Financial Services is considering two plans for raising \$900,000 to expand operations. Plan A is to borrow at 10%, and plan B is to issue 225,000 shares of share capital at \$4.00 per share. Before any new financing, First Bank Financial Services has net income of \$600,000 and 200,000 shares of share capital outstanding. Assume you own most of First Bank Financial Services' existing shares. Its management believes the company can use the new funds to earn additional income of \$800,000 before interest and taxes. First Bank Financial Services' income tax rate is 25%.

Requirements

- Analyze First Bank Financial Services' situation to determine which plan will result in higher earnings per share.
- Which plan results in the higher earnings per share? Which plan allows you to retain control of the company? Which plan creates more financial risk for the company? Which plan do you prefer? Why? Present your conclusion in a memo to First Bank Financial Services' board of directors.

Group B**L0 1**

E9-32B. (Learning Objective 1: Accounting for warranty expense and the related liability) The accounting records of Made of Glass Ceramics included the following balances at the end of the period:

Provision for Warranty Repairs	Warranty Expense
Beg. bal. 4,000	

Made of Glass has determined that its warranty obligations at the end of the year are equal to €12,000. During 20X0, the business paid €5,000 to satisfy the warranty claims.

Requirements

1. Journalize Made of Glass's warranty expense for the period and the company's cash payments to satisfy warranty claims. Explanations are not required.
2. Show what Made of Glass will report on its Income Statement and Balance Sheet for this situation.
3. Which data item from Requirement 2 will affect Made of Glass's current ratio? Will Made of Glass's current ratio increase or decrease as a result of this item?

E9-33B. (*Learning Objective 1: Recording and reporting current liabilities*) Trevor Publishing completed the following transactions for one subscriber during 20X0:

L0 1

Oct 1	Sold a one-year subscription, collecting cash of \$1,400, plus sales tax of 9%.
Nov 15	Remitted (paid) the sales tax to the Limburg Provincial Office.
Dec 31	Made the necessary adjustment at year-end.

Requirement

1. Journalize these transactions (explanations not required). Then report any liability on the company's Balance Sheet at December 31.

E9-34B. (*Learning Objective 1: Reporting payroll expense and liabilities*) MV Talent Search has an annual payroll of €180,000. In addition, the company incurs a payroll tax expense of 9%. At December 31, MV owes salaries of €7,800 and payroll tax of €750. The company will pay these amounts early next year.

L0 1

Show what MV will report for the foregoing on its Income Statement and year-end — Balance Sheet.

E9-35B. (*Learning Objective 1: Recording note payable transactions*) Assume that Hansel Company completed the following note-payable transactions:

L0 1

20X0	
Mar 1	Purchased delivery truck costing \$80,000 by issuing a one-year, 5% note payable.
Dec 31	Accrued interest on the note payable.
20X1	
Mar 1	Paid the note payable at maturity.

Requirements

1. How much interest expense must be accrued at December 31, 20X0? (Round your answer to the nearest whole dollar.)
2. Determine the amount of Hansel's final payment on March 1, 20X1.
3. How much interest expense will Hansel report for 20X0 and for 20X1? (Round your answer to the nearest whole dollar.)

E9-36B. (*Learning Objective 1: Accounting for income tax*) At December 31, 20X0, Sybil Real Estate reported a current liability for income tax payable of €160,000. During 20X1, Sybil earned income of €1,600,000 before income tax. The company's income tax rate during 20X1 was 25%. Also during 20X1, Sybil paid income taxes of €310,000.

L0 1

LO 1 5

■ writing assignment

How much income tax payable did Sybil Real Estate report on its Balance Sheet at December 31, 20X1? How much income tax expense did Saglio report on its 20X1 Income Statement?

E9-37B. (Learning Objectives 1, 5: Analyzing liabilities) Green Earth Structures, Inc., builds environmentally sensitive structures. The company's 20X1 revenues totaled €2,852 million, and at December 31, 20X1, the company had €652 million in current assets. The December 31, 20X1 and 20X0, Balance Sheets reported the liabilities and shareholders' equity as follows.

A1			
	A	B	C
1	At year-end (in millions)	20X1	20X0
2	Liabilities and Shareholders' Equity		
3	Current Liabilities		
4	Accounts payable	€ 145	€ 183
5	Accrued expenses	161	182
6	Employee compensation and benefits	131	15
7	Current portion of long-term debt	3	9
8	Total Current Liabilities	340	389
9	Long-Term Debt	1,488	1,317
10	Post-Retirement Benefits Payable	129	135
11	Other Liabilities	11	7
12	Shareholders' Equity	2,030	1,776
13	Total Liabilities and Shareholders' Equity	€ 3,998	€ 3,624
14			

Requirements

1. Describe each of Green Earth Structures, Inc.'s liabilities and state how the liability arose.
2. What were the company's total assets at December 31, 20X1? Was the company's debt ratio at the end of 20X1 high, low, or in a middle range?

LO 1

E9-38B. (Learning Objective 1: Reporting a contingent liability) Edward Security Systems' revenues for 20X0 totaled €26.7 million. As with most companies, Edward is a defendant in lawsuits related to its products. Note 14 of the Edward Annual Report for 20X0 explains the company's policy on legal proceedings, stating that they will accrue for amounts related to legal matters if liability is probable and if it is reasonable to estimate an amount.

Requirements

1. Suppose Edward's lawyers believe that a significant legal judgment against the company is reasonably possible. How should Edward report this situation in its financial statements?
2. Suppose Edward's lawyers believe it is probable that a €2.3 million judgment will be rendered against the company. Report this situation in Edward's financial statements. Journalize any entry required by IFRS. Explanations are not required.

LO 1 5

E9-39B. (Learning Objectives 1, 5: Reporting current and long-term liabilities) Assume Hi-Tech Electronics completed these transactions during September 20X0.

- a. Sales of €2,150,000 are subject to estimated warranty cost of €112,500. The provision for warranty repairs at the beginning of the year was €34,000, and warranty payments for the year totaled €57,000.
- b. On September 1, Hi-Tech Electronics signed a €44,000 note payable that requires annual payments of €11,000 plus 4% interest on the unpaid balance each September 2.
- c. Music For You, Inc., a chain of music stores, ordered €100,000 worth of CD players. With its order, Music For You, Inc., sent a check for €100,000, and Hi-Tech Electronics

shipped €85,000 of the goods. Hi-Tech Electronics will ship the remainder of the goods on October 3, 20X0.

- d. The September payroll of €250,000 is subject to employee withheld income tax of €30,000 and payroll tax of 7.65%. On September 30, Hi-Tech Electronics pays employees their take-home pay and accrues all tax amounts.

Requirement

1. Report these items on Hi-Tech Electronics' Balance Sheet at September 30, 20X0.

E9-40B. (*Learning Objective 2: Issuing bonds payable [discount]; paying and accruing interest; amortizing the bonds by the effective interest method*) On January 31, Daughtry Logistics, Inc., issued five-year, 5% bonds payable with a face value of €12,000,000. The bonds were issued when the market interest rate was 6% and pay interest on January 31 and July 31. Daughtry Logistics, Inc., amortizes bond discounts by the effective interest method.

L0 2

Requirements

1. Record issuance of the bonds on January 31.
2. Record the semi-annual interest payment and amortization of bond discount on July 31.
3. Record the interest accrual and discount amortization on December 31.

E9-41B. (*Learning Objective 2: Measuring cash amounts for a bond payable [premium]; amortizing the bonds using the straight-line method*) Commonwealth Bank has €400,000 of 9% debenture bonds outstanding. The bonds were issued at 104 in 20X0 and mature in 20Z0 (20 years from now).

L0 2

Requirements

1. How much cash did Commonwealth Bank receive when it issued these bonds?
2. How much cash in total will Commonwealth Bank pay the bondholders through the maturity date of the bonds?
3. Take the difference between your answers to Requirements 1 and 2. What does this number represent?
4. Estimate Commonwealth Bank's annual interest expense by the effective interest amortization method.

E9-42B. (*Learning Objective 2: Issuing bonds payable [discount]; recording interest payments and the related bond amortization*) First Place Sports Ltd. is authorized to issue €1,500,000 of 9%, 10-year bonds payable. On December 31, 20X0, when the market interest rate is 10%, the company issues €840,000 of the bonds. First Place Sports amortizes bonds by the effective-interest method. The semi-annual interest dates are June 30 and December 31.

L0 2

■ spreadsheet

Requirements

1. Prepare a bond amortization table for the first four semi-annual interest periods.
2. Record issuance of the bonds payable on December 31, 20X0, the first semi-annual interest payment on June 30, 20X1, and the second payment on December 31, 20X1.

E9-43B. (*Learning Objective 2: Issuing bonds payable [premium]; recording interest accrual and payment and the related bond amortization*) On June 30, 20X0, the market interest rate is 9%. Team Sports Ltd. issues €3,400,000 of 10%, 10-year bonds payable. The bonds pay interest on June 30 and December 31. Team Sports Ltd. amortizes bonds by the effective-interest method.

L0 2

■ spreadsheet

Requirements

1. Prepare a bond amortization table for the first four semi-annual interest periods.
2. Record the issuance of bonds payable on June 30, 20X0, the payment of interest on December 31, 20X0, and the payment of interest on June 30, 20X1.

■ spreadsheet

LO 2

E9-44B. (*Learning Objective 2: Creating a bond amortization schedule [discount]*) Terry Co. issued €750,000 of 11%, 10-year bonds payable on January 1, 20X0, when the market interest rate was 12%. The company pays interest annually at year-end.

Requirement

1. Create a spreadsheet model to prepare a schedule to amortize the bonds. Use the effective-interest method of amortization. (Round to the nearest dollar.)

LO 3

E9-45B. (*Learning Objective 3: Recording operating and capital lease, preparing lease amortization schedule*) Small Billy Guff entered into a lease agreement for an asset on the following terms:

- Lease period = 4 years
- Lease payment = €18,000 at the end of each period
- Lease interest rate = 8%

Requirements

1. Assuming that this lease qualifies as a capital lease, prepare the lease amortization table, starting from the signing of the lease to the final lease payments. Prepare the journal entry for the first lease payment.
2. Assuming that this lease qualifies as an operating lease, prepare the journal entry for the first lease payment.
3. Explain how a lease for the same item may be treated so differently by two companies.

LO 4

E9-46B. (*Learning Objective 4: Measuring the times-interest-earned ratio*) Companies that operate in different industries may have very different financial ratio values. These differences may grow even wider when we compare companies located in different countries.

Compare three leading companies on their current ratio, debt ratio, and times-interest-earned ratio. Compute three ratios for Company F, Company L, and Company V. Based on your computed ratio values, which company looks the least risky?

A1				
	A	B	C	D
1	(Amounts in millions or billions)	Company F	Company L	Company V
2	Income data			
3	Total revenues	\$ 9,723	¥ 7,312	€ 136,377
4	Operating income	294	229	5,627
5	Interest expense	43	29	687
6	Net income	25	12	443
7				
8	Asset and liability data			
9	(Amounts in millions or billions)			
10	Total current assets	435	5,422	147,398
11	Long-term assets	135	740	61,173
12	Total current liabilities	227	2,248	72,620
13	Long-term liabilities	109	2,320	110,927
14	Shareholders' equity	234	1,594	25,024
15				

LO 4

■ writing assignment

E9-47B. (*Learning Objective 4: Analyzing alternative plans for raising money*) First Federal Financial Services is considering two plans for raising €650,000 to expand operations. Plan A is to borrow at 5%, and plan B is to issue 100,000 shares of share capital at €6.50 per share. Before any new financing, First Federal Financial Services has a net income of €400,000 and 100,000 shares of share capital outstanding. Assume you own most of First Federal Financial Services' existing shares. Its management believes the company can use the new funds to earn additional income of €550,000 before interest and taxes. First Federal Financial Services' income tax rate is 40%.

Requirements

1. Analyze First Federal Financial Services' situation to determine which plan will result in the higher earnings per share.
2. Which plan results in the higher earnings per share? Which plan allows you to retain control of the company? Which plan creates more financial risk for the company? Which plan do you prefer? Why? Present your conclusion in a memo to First Federal Financial Services' board of directors.

Challenge Exercises

E9-48. (Learning Objectives 1, 5: Reporting current liabilities) The top management of Pratt Marketing Services examines the following company accounting records at August 29, immediately before the end of the year, August 31:

L0 1 5

Total current assets	\$ 324,700
Non-current assets	1,067,500
	<u>\$1,392,200</u>
Total current liabilities	\$ 193,400
Non-current liabilities	253,400
Owners' equity	945,400
	<u>\$1,392,200</u>

Suppose Pratt's management wants to achieve a current ratio of 2.5. How much in current liabilities should Pratt pay off within the next two days in order to achieve its goal?

E9-49. (Learning Objectives 2, 5: Refinancing old bonds payable with new bonds) Great Brands completed one of the most famous debt refinancings in history. A debt refinancing occurs when a company issues new bonds payable to retire old bonds. The company debits the old bonds payable and credits the new bonds payable.

L0 2 5

Great Brands had \$150 million of 5.75% bonds payable outstanding, with 21 years to maturity. Great retired these old bonds by issuing \$85 million of new 11% bonds payable to the holders of the old bonds and paying the bondholders \$10 million in cash. Great issued both groups of bonds at face value. At the time of the debt refinancing, Great Brands had total assets of \$497 million and total liabilities of \$357 million. Net income for the most recent year was \$6.2 million on sales of \$1 billion.

Requirements

1. Journalize the debt refinancing transaction.
2. Compute annual interest expense for both the old and the new bond issues.
3. Why did Great Brands refinance the old 5.75% bonds with the new 11% bonds? Consider interest expense, net income, and the debt ratio.

E9-50. (Learning Objectives 2, 3: Analyzing bond transactions) This (adapted) advertisement for a 20-year bond appeared in the *Wall Street Chronicle*.

L0 2 3

■ **writing assignment**

New issue on March 15, 20X0

\$800,000

HOLIDAY CORPORATION

12% Subordinated Debentures due March 15, 20Y0
interest payable March 15 and September 15

Price 94.5%

(Note: A *subordinated debenture* is an unsecured bond payable whose rights are less than the rights of other bondholders.)

Requirements

1. Journalize Holiday's issuance of these bonds payable on March 15, 20X0.
2. Compute the semi-annual cash interest payment on the bonds.
3. Compute the semi-annual interest expense under the effective interest amortization method.
4. Compute both the first-year (from March 15, 20X0, to March 15, 20X1) and the second-year (March 15, 20X1, to March 15, 20X2) interest expense under the effective-interest amortization method. The market rate for similar bonds at the date of issuance was 13%. Why is interest expense greater in the second year?

Quiz

Test your understanding of accounting for liabilities by answering the following questions. Select the best choice from among the possible answers given.

Q9-51. For the purpose of classifying liabilities as current or non-current, the term operating cycle refers to

- a. the time period between date of sale and the date the related revenue is collected.
- b. the average time period between business recessions.
- c. a period of one year.
- d. the time period between purchase of merchandise and the conversion of this merchandise back to cash.

Q9-52. Failure to accrue interest expense results in

- a. an overstatement of net income and an overstatement of liabilities.
- b. an understatement of net income and an overstatement of liabilities.
- c. an understatement of net income and an understatement of liabilities.
- d. an overstatement of net income and an understatement of liabilities.

Q9-53. Feline Warehouse operates in a state with a 5.5% sales tax. For convenience, Feline Warehouse credits Sales Revenue for the total amount (selling price plus sales tax) collected from each customer. If Feline Warehouse fails to make an adjustment for sales taxes,

- a. net income will be overstated and liabilities will be overstated.
- b. net income will be understated and liabilities will be overstated.
- c. net income will be understated and liabilities will be understated.
- d. net income will be overstated and liabilities will be understated.

Q9-54. What kind of account is Unearned Revenue?

- a. Liability account
- b. Asset account
- c. Revenue account
- d. Expense account

Q9-55. An end-of-period adjusting entry that debits Unearned Revenue most likely will credit

- a. a liability.
- b. an asset.
- c. an expense.
- d. a revenue.

Q9-56. Aphrodite, Inc., manufactures and sells computer monitors with a three-year warranty. When calculated using the expected value approach, Aphrodite found that expected warranty costs are roughly equal to about 6% of sales during the warranty period. The following table shows the sales and actual warranty payments during the first two years of operations:

Year	Sales	Warranty Payments
20X0	\$450,000	\$ 3,150
20X1	750,000	30,000

Based on these facts, what amount of warranty liability should Aphrodite, Inc., report on its Balance Sheet at December 31, 20X1?

- a. \$30,000
- b. \$33,150
- c. \$38,850
- d. \$72,000

Q9-57. Tomorrow's Fashions has a debt that has been properly reported as a long-term liability up to the present year (20X0). Some of this debt comes due in 20X0. If Tomorrow's Fashions continues to report the current position as a long-term liability, the effect will be to

- a. overstate the current ratio.
- b. understate total liabilities.
- c. overstate net income.
- d. understate the debt ratio.

Q9-58. A bond with a face amount of \$10,000 has a current price quote of 103.625. What is the bond's price?

- a. \$1,036.25
- b. \$10,362.50
- c. \$10,103.63
- d. \$1,036,250

Q9-59. Bond carrying value equals Bonds Payable

- a. plus Premium on Bonds Payable.
- b. plus Discount on Bonds Payable.
- c. minus Discount on Bonds Payable.
- d. minus Premium on Bonds Payable.
- e. Both a and c
- f. Both b and d

Q9-60. What type of account is Discount on Bonds Payable and what is its normal balance?

- a. Adjusting amount; Credit
- b. Reversing account; Debit
- c. Contra liability; Debit
- d. Contra liability; Credit

Questions 61–64 use the following data:

Spring Company sells \$200,000 of 12%, 10-year bonds with \$192,000 on April 1, 20X0. The market rate of interest on that day is 12.5%. Interest is paid each year on April 1.

Q9-61. The entry to record the sale of the bonds on April 1 would be

A1				
	A	B	C	D
1	a.	Cash	200,000	
2		Bonds Payable		200,000
3				
4	b.	Cash	192,000	
5		Discount on Bonds Payable	8,000	
6		Bonds Payable		200,000
7				
8	c.	Cash	192,000	
9		Bonds Payable		192,000
10				
11	d.	Cash	200,000	
12		Discount on Bonds Payable		8,000
13		Bonds Payable		192,000
14				

Q9-62. Spring Company uses the effective interest amortization method. The amount of interest expense on April 1 of each year will be

- a. \$24,000.
- b. \$24,800.
- c. \$25,000.
- d. \$32,000.
- e. none of these.

Q9-63. Write the adjusting entry required at December 31, 20X0.

Q9-64. Write the journal entry requirements at April 1, 20X1.

Q9-65. McPartlin Corporation issued \$350,000 of 10%, 10-year bonds payable on January 1, 20X0, for \$275,695. The market interest rate when the bonds were issued was 14%. Interest is

paid semi-annually on January 1 and July 1. The first interest payment is July 1, 20X0. Using the effective-interest amortization method, how much interest expense will McPartlin record on July 1, 20X0?

- a. \$17,500
- b. \$10,500
- c. \$24,500
- d. \$19,425
- e. \$19,299

Q9-66. Using the facts in the preceding question, McPartlin's journal entry to record the interest expense on July 1, 20X0, will include a

- a. credit to Discount on Bonds Payable.
- b. debit to Bonds Payable.
- c. debit to Premium on Bonds Payable.
- d. credit to Interest Expense.

Q9-67. Amortizing the discount on bonds payable

- a. is necessary only if the bonds were issued at more than face value.
- b. increases the recorded amount of interest expense.
- c. reduces the semi-annual cash payment for interest.
- d. reduces the carrying value of the bond liability.

Q9-68. The journal entry on the maturity date to record the payment of \$600,000 of bonds payable that were issued at a \$60,000 discount includes

- a. a debit to Discount on Bonds Payable for \$60,000.
- b. a credit to Cash for \$660,000.
- c. a debit to Bonds Payable for \$600,000.
- d. all of the above.

Q9-69. Is the payment of the face amount of a bond on its maturity date regarded as an operating activity, an investing activity, or a financing activity?

- a. Operating activity
- b. Investing activity
- c. Financing activity

Problems MyLab Accounting

Select A and B problems can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select problems using MyLab Accounting.

Group A

L0 1

P9-70A. (Learning Objective 1: Measuring current liabilities) Deep Waters Marine experienced these events during the current year:

- a. Its December revenue totaled \$140,000, and Deep Waters collected sales tax of 5%. The tax amount will be sent to the Hanoi Municipality early in January.
- b. On August 31, Deep Waters signed a six-month, 4% note payable to purchase a boat costing \$87,000. The note requires payment of principal and interest at maturity.
- c. On August 31, Deep Waters received cash of \$2,700 in advance for service revenue. This revenue will be earned evenly over six months.
- d. Revenues of \$860,000 were covered by Deep Waters's service warranty. At January 1, provision for warranty repairs was \$11,800. During the year, Deep Waters recorded warranty expense of \$34,400 and paid warranty claims of \$34,800.
- e. Deep Waters owes \$60,000 on a long-term note payable. At December 31, 12% interest for the year plus \$30,000 of this principal are payable within one year.

Requirement

1. For each item, indicate the account and the related amount to be reported as a current liability on the Deep Waters Marine Balance Sheet at December 31.

L0 1

P9-71A. (Learning Objective 1: Recording liability-related transactions) The following transactions of Sweet Ensemble Music Company occurred during 20X0 and 20X1:

20X0		
May 3		Purchased a piano (inventory) for \$72,000, signing a six-month, 4% note payable.
May 31		Borrowed \$75,000 on a 5% note payable that calls for annual installment payments of \$15,000 principal plus interest. Recorded the short-term note payable in a separate account from the long-term note payable.
Sep 3		Paid the six-month, 4% note at maturity.
Dec 31		Accrued warranty expense, which is estimated at 3.0% of sales of \$192,000.
	31	Accrued interest on the outstanding note payable.
20X1		
May 31		Paid the first installment and interest for one year on the outstanding note payable.

Requirement

- Record the transactions in Sweet Ensemble's journal. Explanations are not required.

P9-72A. (Learning Objective 2: Recording bond transactions [at par]; reporting bonds payable on the Balance Sheet) The board of directors of Displays Plus authorizes the issue of \$8,000,000 of 10%, five-year bonds payable. The semi-annual interest dates are May 31 and November 30. The bonds are issued on May 31, 20X0, at par.

L0 2

Requirements

- Journalize the following transactions:
 - Issuance of half of the bonds on May 31, 20X0.
 - Payment of interest on November 30, 20X0.
 - Accrual of interest on December 31, 20X0.
 - Payment of interest on May 31, 20X1.
- Report interest payable and bonds payable as they would appear on the Displays Plus Balance Sheet at December 31, 20X0.

P9-73A. (Learning Objectives 2, 5: Issuing bonds at a premium; amortizing by the effective interest method; reporting bonds payable on the Balance Sheet) On February 28, 20X0, Nemo Corp. issues 8%, 10-year bonds payable with a face value of \$800,000. The bonds pay interest on February 28 and August 31. Nemo Corp. amortizes bonds by the effective interest method.

L0 2 5

Requirements

- If the market interest rate is 7% when Nemo Corp. issues its bonds, will the bonds be priced at par, at a premium, or at a discount? Explain.
- If the market interest rate is 9% when Nemo Corp. issues its bonds, will the bonds be priced at par, at a premium, or at a discount? Explain.
- Assuming that the market rate is 7%, journalize the following bonds payable transactions.
 - Issuance of the bonds on February 28, 20X0.
 - Payment of interest and amortization of the bonds on August 31, 20X0.
 - Accrual of interest and amortization of the bonds on December 31, 20X0.
 - Payment of interest and amortization of the bonds on February 28, 20X1.
- Report interest payable and bonds payable as they would appear on the Nemo Corp. Balance Sheet at December 31, 20X0.

P9-74A. (Learning Objective 2: Accounting for bonds payable at a premium; amortizing by the effective interest method)

L0 2

Requirements

- Journalize the following transactions of Dupont Communications, Inc.:

20X0		
Jan	1	Issued \$6,000,000 of 9%, 10-year bonds payable when the market rate is 8%.
Jul	1	Paid semi-annual interest and amortized bonds by the effective interest method on the 9% bonds payable.
Dec	31	Accrued semi-annual interest expense and amortized bonds by the effective interest method on the 9% bonds payable.
20X1		
Jan	1	Paid semi-annual interest.
2020		
Jan	1	Paid the 9% bonds at maturity.

2. At December 31, 20X0, after all year-end adjustments, determine the carrying amount of Dupont Communications bonds payable, net.
3. For the six months ended July 1, 20X0, determine for Dupont Communications, Inc.:
 - a. Interest expense
 - b. Cash interest paid

What causes interest paid for the bonds to be higher than the interest expense?

L0 2 5

■ spreadsheet

P9-75A. (*Learning Objectives 2, 5: Analyzing a company's long-term debt; reporting long-term debt on the Balance Sheet [effective-interest method]*) The notes to the Helping Charities' financial statements reported the following data on December 31, Year 1 (end of the fiscal year):

Note 6. Indebtedness		
Bonds payable, 7% due in Year 7	\$3,500,000	
Less: Discount	(162,365)	\$3,337,635
Notes payable, 6%, payable in amounts of \$55,000 annual installments starting in Year 5.....		330,000

Helping Charities amortizes bonds by the effective-interest method and pays all interest amounts at December 31.

Requirements

1. Answer the following questions about Helping Charities' long-term liabilities:
 - a. What is the maturity value of the 7% bonds?
 - b. What are Helping Charities' annual cash interest payments on the 7% bonds?
 - c. What is the carrying amount of the 7% bonds at December 31, year 1?
2. Prepare an amortization table through December 31, Year 4, for the 7% bonds. The market interest rate on the bonds was 8%. (Round all amounts to the nearest dollar.) How much is Helping Charities' interest expense on the 7% bonds for the year ended December 31, Year 4?
3. Show how Helping Charities would report the 7% bonds payable and the 6% notes payable at December 31, Year 4.

L0 3

■ spreadsheet

P9-76A. (*Learning Objective 3: Determining appropriate lease category, journalizing lease transactions*) On December 31, 20X0, Meekaboo Corp. entered into the following lease for an asset with six years' useful life and a fair market value of \$180,000.

- Lease period = 4 years
- Lease payment of \$55,000 per year at the end of each year
- Interest rate implicit in the lease = 10%

Requirements

1. Based on the lease terms, determine if Meekaboo should record an operating or finance lease.
2. Journalize all entries related to the lease between December 31, 20X0 and 20X1.

P9-77A. (Learning Objective 4: Financing operations with debt or with shares) Viola Sporting Goods is embarking on a massive expansion. Assume plans call for opening 25 new stores during the next three years. Each store is scheduled to be 30% larger than the company's existing locations, offering more items of inventory, and with more elaborate displays. Its management estimates that company operations will provide \$1.5 million of the cash needed for expansion. Viola must raise the remaining \$7.5 million from outsiders. The board of directors is considering obtaining the \$7.5 million either through borrowing or by issuing share capital.

Requirement

1. Write a memo to Viola's management discussing the advantages and disadvantages of borrowing and of issuing share capital to raise the needed cash. Which method of raising the funds would you recommend?

P9-78A. (Learning Objectives 4, 5: Reporting liabilities on the Balance Sheet; calculating the times-interest-earned ratio) The accounting records of Quinzel Foods, Inc., include the following items at December 31, 20X0:

Mortgage note payable, current	\$ 94,000	Accumulated depreciation, equipment	\$164,000
Accumulated pension benefit obligation	465,000	Discount on bonds payable (all long-term)	27,000
Bonds payable, long-term	1,200,000	Operating income	400,000
Mortgage note payable, long-term	319,000	Equipment	745,000
Bonds payable, current portion ...	400,000	Pension plan assets (market value)	405,000
Interest expense	222,000	Interest payable	72,000

Requirements

1. Show how each relevant item would be reported on the Quinzel Foods, Inc., classified Balance Sheet, including headings and totals for current liabilities and long-term liabilities.
2. Answer the following questions about Quinzel's financial position at December 31, 20X0:
 - a. What is the carrying amount of the bonds payable (combine the current and long-term amounts)?
 - b. Why is the interest-payable amount so much less than the amount of interest expense?
3. How many times did Quinzel cover its interest expense during 20X0?

Group B

P9-79B. (Learning Objective 1: Measuring current liabilities) Tropical Shore Marine experienced these events during the current year.

- a. December revenue totaled €120,000, and Tropical Shore collected sales tax of 8%. The tax amount will be sent to the Groningen provincial office early in January.
- b. On August 31, Tropical Shore signed a six-month, 4% note payable to purchase a boat costing €81,000. The note requires payment of principal and interest at maturity.
- c. On August 31, Tropical Shore received cash of €1,500 in advance for service revenue. This revenue will be earned evenly over six months.
- d. Revenues of €780,000 were covered by Tropical Shore's service warranty. At January 1, provision for warranty repairs was €11,400. During the year, Tropical Shore recorded warranty expense of €31,200 and paid warranty claims of €34,600.
- e. Tropical Shore owes €88,000 on a long-term note payable. At December 31, 10% interest for the year plus €24,000 of this principal are payable within one year.

Requirement

1. For each item, indicate the account and the related amount to be reported as a current liability on the Tropical Shore Marine Balance Sheet at December 31.

LO 4

■ writing assignment

LO 4 5

LO 1

L0 1

P9-80B. (*Learning Objective 1: Recording liability-related transactions*) The following transactions of Lovely Melody Music Company occurred during 20X0 and 20X1:

20X0		
Mar 3	Purchased a piano (inventory) for €35,000, signing a six-month, 10% note payable.	
May 31	Borrowed €75,000 on an 8% note payable that calls for annual installment payments of €15,000 principal plus interest. Recorded the short-term note payable in a separate account from the long-term note payable.	
Sep 3	Paid the six-month, 10% note at maturity.	
Dec 31	Accrued warranty expense, which is estimated at 1.5% of sales of €198,000.	
31	Accrued interest on the outstanding note payable.	
20X1		
May 31	Paid the first installment and interest for one year on the outstanding note payable.	

Requirement

- Record the transactions in Lovely Melody Music Company's journal. Explanations are not required.

L0 2

P9-81B. (*Learning Objective 2: Recording bond transactions [at par]; reporting bonds payable on the Balance Sheet*) The board of directors of Portraits Plus authorizes the issue of €4,000,000 of 8%, 15-year bonds payable. The semi-annual interest dates are May 31 and November 30. The bonds are issued on May 31, 20X0, at par.

Requirements

- Journalize the following transactions:
 - Issuance of half of the bonds on May 31, 20X0.
 - Payment of interest on November 30, 20X0.
 - Accrual of interest on December 31, 20X0.
 - Payment of interest on May 31, 20X1.
- Report the interest payable and bonds payable as they would appear on the Portraits Plus Balance Sheet at December 31, 20X0.

L0 2 5

P9-82B. (*Learning Objectives 2, 5: Issuing bonds at a premium; amortizing by the effective interest method; reporting notes payable on the Balance Sheet*) On February 28, 20X0, Dory Corp. issues 6%, 20-year bonds payable with a face value of €1,500,000. The bonds pay interest on February 28 and August 31. Dory Corp. amortizes bonds by the effective interest method.

Requirements

- If the market interest rate is 5% when Dory Corp. issues its bonds, will the bonds be priced at par, at a premium, or at a discount? Explain.
- If the market interest rate is 7% when Dory Corp. issues its bonds, will the bonds be priced at par, at a premium, or at a discount? Explain.
- Assuming the market rate is 5%, journalize the following bond transactions:
 - Issuance of the bonds on February 28, 20X0.
 - Payment of interest and amortization of the bonds on August 31, 20X0.
 - Accrual of interest and amortization of the bonds on December 31, 20X0, the year-end.
 - Payment of interest and amortization of the bonds on February 28, 20X1.
- Report interest payable and bonds payable as they would appear on the Dory Corp. Balance Sheet at December 31, 20X0.

L0 2

P9-83B. (*Learning Objective 2: Accounting for bonds payable at a premium; amortizing by the effective interest method*)

Requirements

1. Journalize the following transactions of Seville Communications, Inc.:

20X0		
Jan	1	Issued \$2,000,000 of 7%, 10-year bonds payable when the market rate is 6%.
Jul	1	Paid semi-annual interest and amortized the bonds by the effective interest method on the 7% bonds payable.
Dec	31	Accrued semi-annual interest expense and amortized the bonds by the effective interest method on the 7% bonds payable.
20X1		
Jan	1	Paid semi-annual interest.
2020		
Jan	1	Paid the 7% bonds at maturity.

2. At December 31, 20X0, after all year-end adjustments, determine the carrying amount of Seville Communications bonds payable, net.
3. For the six months ended July 1, 20X0, determine the following for Seville Communications, Inc:
 - a. Interest expense
 - b. Cash interest paid

What causes the interest paid for the bonds to be higher than the interest expense?

P9-84B. (*Learning Objectives 2, 5: Analyzing a company's long-term debt; reporting the long-term debt on the Balance Sheet [effective-interest method]*) The notes to the Helpful Charities financial statements reported the following data on December 31, Year 1 (end of the fiscal year):

L0 2 5
■ spreadsheet

Note 6. Indebtedness		
Bonds payable, 4% due in Year 7	€6,500,000	
Less: Discount	(331,240)	€6,168,760
Notes payable, 7%, payable in €60,000 annual installments starting in Year 5		360,000

Helpful Charities amortizes bonds by the effective-interest method and pays all interest amounts at December 31.

Requirements

1. Answer the following questions about Helpful Charities' long-term liabilities:
 - a. What is the maturity value of the 4% bonds?
 - b. What is Helpful Charities' annual cash interest payment on the 4% bonds?
 - c. What is the carrying amount of the 4% bonds at December 31, Year 1?
2. Prepare an amortization table through December 31, Year 4, for the 4% bonds. The market interest rate on the bonds was 5%. Round all amounts to the nearest dollar. How much is Helpful Charities' interest expense on the 4% bonds for the year ended December 31, Year 4?
3. Show how Helpful Charities would report the 4% bonds and the 7% notes payable at December 31, Year 4.

P9-85B. (*Learning Objective 3: Determining appropriate lease category, journalizing lease transactions*) On December 31, 20X0, Reekaboo Corp. entered into the following lease for an asset with 8 years useful life and a fair market value of €160,000.

- Lease period = 6 years
- Lease payment of €34,000 per year at the end of each year
- Interest rate implicit in the lease = 8%

L0 3
■ spreadsheet

Requirements

1. Based on the lease terms, determine if Reekaboo should record an operating or finance lease.
2. Journalize all entries related to the lease between December 31, 20X0 and 20X1.

L0 4

■ writing assignment

P9-86B. (Learning Objective 4: Financing operations with debt or with shares) Veronica Sporting Goods is embarking on a massive expansion. Assume plans call for opening 30 new stores during the next four years. Each store is scheduled to be 35% larger than the company's existing locations, offering more items of inventory, and with more elaborate displays. Its management estimates that company operations will provide €1.75 million of the cash needed for expansion. Veronica must raise the remaining €8.25 million from outsiders. The board of directors is considering obtaining the €8.25 million either through borrowing or by issuing share capital.

Requirement

1. Write a memo to Veronica's management discussing the advantages and disadvantages of borrowing and of issuing share capital to raise the needed cash. Which method of raising the funds would you recommend?

L0 4 5

P9-87B. (Learning Objectives 4, 5: Reporting liabilities on the Balance Sheet; calculating the times-interest-earned ratio) The accounting records of Isley Foods, Inc., include the following items at December 31, 20X0:

Mortgage note payable, current	€ 95,000	Accumulated depreciation, equipment	€165,000
Accumulated pension benefit obligation	460,000	Discount on bonds payable (all long-term)	23,000
Bonds payable, long-term	200,000	Operating income	360,000
Mortgage note payable, long-term	313,000	Equipment	746,000
Bonds payable, current portion ...	500,000	Pension plan assets (market value)	410,000
Interest expense	224,000	Interest payable	72,000

Requirements

1. Show how each relevant item would be reported on the Isley Foods, Inc., classified Balance Sheet, including headings and totals for current liabilities and long-term liabilities.
2. Answer the following questions about Isley's financial position at December 31, 20X0:
 - a. What is the carrying amount of the bonds payable (combine the current and long-term amounts)?
 - b. Why is the interest-payable amount so much less than the amount of interest expense?
3. How many times did Isley cover its interest expense during 20X0?

APPLY YOUR KNOWLEDGE

Decision Cases

L0 2

Case 1. (Learning Objective 2: Exploring an actual bankruptcy) In 2002, Enron Corporation filed for Chapter 11 bankruptcy protection, shocking the business community: How could a company this large and this successful go bankrupt? This case explores the causes and the effects of Enron's bankruptcy.

At December 31, 2000, and for the four years ended on that date, Enron reported the following (amounts in millions):

Balance Sheet (summarized)				
Total assets				\$65,503
Total liabilities				54,033
Total shareholders' equity				11,470
Income Statements (excerpts)				
	2000	1999	1998	1997
Net income	\$979*	\$893	\$703	\$105

*Operating income = \$1,953
Interest expense = \$838

Unknown to investors and lenders, Enron also controlled hundreds of partnerships that owed vast amounts of money. These special-purpose entities (SPEs) did not appear on the Enron financial statements. Assume that the SPEs' assets totaled \$7,500 million and their liabilities stood at \$7,300 million; assume a 10% interest rate on these liabilities.

During the four-year period up to December 31, 2000, Enron's share price shot up from \$17.50 to \$90.56. Enron used its escalating share price to finance the purchase of the SPEs by guaranteeing lenders that Enron would give them Enron shares if the SPEs could not pay their loans.

In 2001, the SEC launched an investigation into Enron's accounting practices. It was alleged that Enron should have been including the SPEs in its financial statements all along. Enron then restated net income for the years up to 2000, wiping out nearly \$800 million of total net income (and total assets) for this four-year period. Enron's share price tumbled, and the guarantees to the SPEs' lenders added millions to Enron's liabilities (assume the full amount of the SPEs' debt). To make matters worse, the assets of the SPEs lost much of their value; assume that their market value is only \$500 million.

Requirements

1. Compute the debt ratio that Enron reported at the end of 2000. Recompute this ratio after including the SPEs in Enron's financial statements. Also compute Enron's times-interest-earned ratio both ways for 2000. Assume that the changes to Enron's financial position occurred during 2000.
2. Why does it appear that Enron failed to include the SPEs in its financial statements? How do you view Enron after including the SPEs in the company's financial statements? (Challenge)

Case 2. (Learning Objective 4: Analyzing alternative ways of raising \$6 million) Business is going well for Park 'N Fly, the company that operates remote parking lots near major airports. The board of directors of this family-owned company believes that Park 'N Fly could earn an additional \$1.5 million income before interest and taxes by expanding into new markets. However, the \$6 million that the business needs for growth cannot be raised within the family. The directors, who strongly wish to retain family control of the company, must consider issuing securities to outsiders. The directors are considering three financing plans.

Plan A is to borrow at 6%. Plan B is to issue 100,000 shares of share capital. Plan C is to issue 100,000 shares of non-voting, \$3.75 preference shares (\$3.75 is the annual dividend paid on each share of preference shares). Park 'N Fly presently has net income of \$3.5 million and 1 million shares of share capital outstanding. The company's income tax rate is 35%.

Requirements

1. Prepare an analysis to determine which plan will result in the highest earnings per share of share capital.
2. Recommend a plan to the board of directors. Give your reasons.

L0 4



Ethical Issues

Issue 1. Microsoft Corporation is the defendant in numerous lawsuits claiming unfair trade practices. Microsoft has strong incentives not to disclose these contingent liabilities. However, accounting standards require that companies report their contingent liabilities.

Requirements

1. Why would a company prefer not to disclose its contingent liabilities?
2. Identify the parties involved in the decision and the potential consequences to each.
3. Analyze the issue of whether to report contingent liabilities from lawsuits from the following standpoints:
 - a. economic
 - b. legal
 - c. ethical

Issue 2. When is a lease a capital idea? Laurie Gocker, Inc., entered into a lease arrangement with Nathan Morgan Leasing Corporation for an industrial machine. Morgan's primary business is leasing. The cash purchase price of the machine is \$1,000,000. Its economic life is six years.

Gocker's Balance Sheet reflects total assets of \$10 million and total liabilities of \$7.5 million. Among the liabilities is a \$2.5 million long-term note outstanding at Last National Bank. The note carries a restrictive covenant that requires the company's debt ratio to be no higher than 75%. The company's revenues have been falling of late and the shareholders are concerned about profitability.

Gocker and Morgan are engaging in negotiations for terms of the lease. Some relevant other facts:

1. Morgan wants to take possession of the machine at the end of the initial lease term.
2. The term may run from four to five years, at Gocker's discretion.
3. Morgan estimates the machine will have no residual value, and Gocker will not purchase it at the end of the lease term.
4. The present value of minimum lease payments on the machine is \$890,000.

Requirements

1. What is (are) the ethical issue(s) in this case?
2. Who are the stakeholders? Analyze the consequences for each stakeholder from the following standpoints: (a) economic, (b) legal, and (c) ethical.
3. How should Gocker structure the lease agreement?

Focus on Financials: | Nestlé



This case spans all 12 chapters and is based on the consolidated financial statements of Nestlé Corporation. As you work with Nestlé throughout this course, you will develop the confidence and ability to use the financial statements of other companies as well. Refer to Nestlé's financial statements in Appendix A (available on www.pearsonglobaleditions.com/Harrison). If you wish, you can obtain the full annual report from www.nestle.com/investors. You may find the information overwhelming for now, but try to spot the key principles that we have discussed in this chapter. It will get progressively easier as you gain familiarity with the elements of the financial statements.

Refer to Nestlé's consolidated financial statements in Appendix A.

1. Did trade payables for Nestlé increase or decrease in 2011? What was the amount? What might have caused this change? Refer to note 12.
2. Examine Note 13—Taxation in the Notes to the Consolidated Financial Statements. What was Nestlé's income tax expense in 2016?
3. Did Nestlé borrow more or pay off more long-term debt during 2016? How can you tell?
4. Examine Note 11—Provisions and Contingencies Liabilities in the Notes to Consolidated Financial Statements. Describe some of Nestlé's contingent liabilities as of March 31, 2011.
5. How would you rate Nestlé's overall debt position—risky, safe, or average? Compute the ratio(s) at December 31, 2016, that would help you answer this question. Ratios aside, what qualitative factors would you consider in assessing the riskiness of Nestlé's overall debt?
6. Refer to Note 11—Provisions and Contingencies. What kind of provisions does Nestlé have? Why does it need them? What do you understand by the term "provision," and how is it different from "contingent liabilities?"

Group Projects

Project 1. Consider three different businesses:

1. A bank
2. A magazine publisher
3. A department store

For each business, list all of its liabilities—both current and long-term. Then compare the three lists to identify the liabilities that the three businesses have in common. Also identify the liabilities that are unique to each type of business.

Project 2. Alcenon Corporation leases the majority of the assets that it uses in operations. Alcenon prefers operating leases (versus capital leases) in order to keep the lease liability off its Balance Sheet and maintain a low debt ratio.

Alcenon is negotiating a 10-year lease on an asset with an expected useful life of 15 years. The lease requires Alcenon to make 10 annual lease payments of \$20,000 each, with the first payment due at the beginning of the lease term. The leased asset has a market value of \$135,180. The lease agreement specifies no transfer of title to the lessee and includes no bargain purchase option.

Write a report for Alcenon's management to explain what conditions must be present for Alcenon to be able to account for this lease as an operating lease.

Quick Check Answers

1. *b*
2. *e*
3. *a* ($\$472,500 \times 0.01 = \$4,725$)
4. *b* [$800 \times 0.04 \times 25 =$ warranty expense of \$800; repaired $\$25 \times 15 = \375 ; year-end liability = \$425 ($\$800 - \375)]
5. *d*
6. *e*
7. *b*
8. *a*
9. *c*
10. *c*
11. *d*
- 12–16.

	Interest Payment	Interest Expense	Discount Amortiz.	Bond Carry- Amt.
1/1/2011				\$521,724
7/1/2011	\$19,600	\$20,869	\$1,269	522,993
1/1/2012	19,600	20,920	1,320	524,313

12. *d* ($\$521,724 \times 0.08 \times 6/12 = \$20,869$)
13. *b* [Int. exp. = \$20,869 Int. payment = \$19,600 ($\$560,000 \times 0.07 \times 6/12$) $\$20,869 - \$19,600 = \$1,269$]
14. *c* ($\$560,000 \times 0.07 = \$39,200$)
15. *a* ($\$20,869 + \$20,920 = \$41,789$)
16. *a* (See Amortization Schedule)

MyLab Accounting

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